

Generative AI and Labor Markets: Wages, Informality, and Causal Evidence from Latin America*

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Abstract

This paper provides the first causal estimates of the effect of large language models (LLMs) on employment and wages outside the OECD: a continuous-treatment difference-in-differences design, estimated country by country on harmonized labor force surveys from eight Latin American countries (2021–2025), with the occupational exposure score of Eloundou et al. as the dose, validated against observed Claude usage. Aggregate effects are heterogeneous and fragile, and employment responds little. The systematic signal is a price: building the outcomes separately for formal and informal workers, exposure lowers the formal monthly wage in six of the eight countries, five of which clear a pre-COVID placebo run on the same sector cells, while informal wages rise in Brazil, Colombia and Ecuador and formal employment contracts in Uruguay. An individual-level composition test locates the adjustment at the entry margin: the entrant-incumbent wage differential is negative and significant in all five countries with tenure data, and in Peru and Brazil incumbent earnings do not move at all. A placebo that rebuilds the design on 2015–2019 with a fictitious launch validates the pattern, and a formal heterogeneity test confirms that the cross-country differences exceed sampling noise. The formal-informal decomposition, which the high-income economies where the existing causal estimates were produced cannot support because their informal sectors are marginal, is the central contribution.

JEL Codes: J23, J31, O33, O54, D63, L86

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1 Introduction

Generative artificial intelligence is widely expected to be the defining shock of the 2020s labor markets. The launch of ChatGPT on November 30, 2022 put the first general-purpose large language model (LLM) in the hands of hundreds of millions of users within months, and occupational exposure measures suggest that [around half of the tasks in US employment could be completed in half the time with software built on LLMs \(Eloundou et al., 2024\)](#). Against these projections, the first causal estimates are striking. Humlum and Vestergaard (2025a) use Danish administrative data linked to adoption surveys sampling more than 100,000 workers and rule out effects on wages and hours larger than two percent two years after the launch. Such effects are small, and the evidence is geographically narrow: nearly every credible causal estimate of how LLMs have moved aggregate labor markets comes from the United States, Denmark, or online freelance platforms dominated by workers from advanced economies.¹

The causal evidence splits into two planes that should not be conflated. At the firm and task level, controlled trials document substantial productivity gains: 14 percent in a customer service center (Brynjolfsson, Li, & Raymond, 2023) and 26 percent more programming tasks completed (Cui et al., 2025); outside the laboratory, adoption of pre-generative predictive AI in French firms shows productivity dominating displacement (Aghion et al., 2025), with no detectable footprint on wage inequality across occupations in the OECD (Georgieff, 2024). At the aggregate labor market level the evidence is mixed: wage gains without employment effects in the United States (Chen, Kane, et al., 2025; Hartley et al., 2026), demand declines of 2 to 5 percent on online labor platforms (Hui et al., 2024) and of 22 to 35 percent (Liu, Xu, et al., 2025), contraction of posted vacancies (Liu, Wang, & Yu, 2025), falling demand in structured cognitive tasks alongside expansion in human-AI collaboration jobs (Chen, Srinivasan, & Zakerinia, 2024; Marguerit, 2025), employment declines concentrated among entry-level workers (Brynjolfsson, Chandar, & Chen, 2025; Klein Teeselink, 2025), and the Danish null. The tension between these planes deserves a precise statement: controlled trials measure within-firm productivity gains under exogenously delivered adoption, not labor market equilibrium outcomes; and the quasi-experimental evidence on that equilibrium is not uniformly null but heterogeneous. The unresolved question is twofold: how much of the within-firm gain reaches labor market aggregates, and why that transmission differs so much across institutional contexts and margins of adjustment. This article addresses the second term.

The shock has reached the region. Weekly Google Trends search interest for “ChatGPT” in the sample countries is indistinguishable from zero during the two months prior to the launch, rises sharply throughout 2023, and stabilizes between forty and ninety on the within-country normalized index (the full series is in the replication package). Whether this awareness has moved labor market outcomes is a distinct empirical question, and for developing countries an unresolved one. Latin America and the Caribbean (LAC) is the

¹Goldsmith-Pinkham (2024) documents the growth of the DiD literature and its use for this type of question: the fraction of NBER applied microeconomics *papers* mentioning DiD or event studies rose from approximately 10 percent in 2002 to more than 30 percent in 2024.

natural place to pose it. The region’s aggregate informality rate stands near 55 percent, ranging from approximately 25 percent in Uruguay to more than 70 percent in parts of the Andean region. Formal and informal workers occupy segmented markets with distinct wage-setting institutions, exposure to digital technology, and buffers against productivity shocks. In the harmonized panels the margin is first order: baseline informality ranges from 23.4 percent (Uruguay) to 73.2 percent (Ecuador) and, in all eight countries, is lower in the occupations most exposed to LLMs than in the least exposed (Table 3), so the shock lands differently on each segment. LLM exposure in the region also differs structurally from the OECD’s: Azuara Herrera et al. (2024) show that predicted exposure tilts toward formal, female, higher-income, and tertiary-educated workers in Chile, Mexico, and Peru, and Ciaschi et al. (2025) and Egaña-delSol and Bravo-Ortega (2026) extend the picture to the region as a whole. No existing article provides a causal estimate of whether that exposure has moved labor market outcomes.

This article fills that gap. I estimate how LLM exposure has affected employment, hours worked, and wages in the region, and whether the effect differs between the formal and informal sectors. I build harmonized labor surveys for eight Latin American countries (Costa Rica, Colombia, Ecuador, Mexico, Peru, Uruguay, Brazil, and Bolivia) covering 2021Q1 to 2025Q4; the last two use documented national adaptations of ISCO-08 (Section 4.1). I map the complementary exposure score β (the share of each occupation’s tasks whose execution time can be at least halved using software built on LLMs) of Eloundou et al. (2024) from O*NET to ISCO-08² and estimate its effect with a continuous-treatment difference-in-differences design, following Callaway, Goodman-Bacon, and Sant’Anna (2025): under their strong parallel trends assumption, each country’s coefficient is a weighted average of the average causal responses (ACRT) along the continuous distribution of exposure (Section 3.3). The strategy is executed *country by country*: each of the eight countries delivers its own estimate $\hat{\tau}^c$ and its own decomposition by baseline formality, and the main result is not a pooled coefficient but the side-by-side comparison of the eight countries and, within each of them, the decomposition of the effect between formal and informal workers, which is where the systematic signal is concentrated; the pre-trend tests and the Honest-DiD breakdown bounds are reported country by country in the results section. A single global launch date may seem thin for identification, but the temporal structure is unusually clean: the November 2022 launch was a worldwide event dated identically in every country³, so the country-by-country regressions share the treatment date $t^* = 2022Q4$ (coded as pre-treatment because two thirds of it precede the launch; treatment is effective from 2023Q1 onward). The choice of β over the alternatives α and ζ from the same article is justified in Section 3.2: β captures diffusion via software built on LLMs, the dominant adoption channel in the region.

The answer is a price rather than a quantity. Aggregate responses are heterogeneous and, in half the sample, indistinguishable from zero. Once the outcome is built separately for formal and informal workers, however, exposure lowers the formal monthly wage in six

²The mapping’s match rate is 98.2 percent of four-digit ISCO-08 codes with positive employment; Section 3.2 details the procedure.

³ChatGPT did not arrive alone: competing general-purpose LLMs entered the market in the following months (Google’s Gemini, Anthropic’s Claude, DeepSeek’s R1), deepening the same channel rather than opening a new one. The relevant adoption date for identification remains November 2022.

of the eight countries, and in five of those six the decline survives a placebo that rebuilds the entire design on the 2015–2019 vintages with a fictitious launch; Mexico is the exception, where the fictitious launch reproduces a larger decline than the real one, so it is reported but not counted. Employment moves far less: the formal expansion holds only in Peru and, marginally, in Costa Rica, and Uruguay reverses it. An individual-level composition test then locates the adjustment: the entrant-incumbent wage differential is negative and significant in all five countries with tenure data, and where contracts protect the established, in Peru and Brazil, their earnings do not move at all. The shock lands on the price at which formal jobs are filled, not on the price paid to those who already hold them.

The post-COVID window is deliberate: including periods prior to 2020 would pool labor markets in two structurally distinct equilibria, violating the comparability requirement for the DiD control group. The launch of ChatGPT delivers an adoption date common to all occupations within a country; identification comes from within-occupation variation in how exposure interacts with the post-launch indicator. The central heterogeneity margin is baseline (2021) formality, not contemporaneous formality, itself an outcome of LLM exposure and therefore a bad control.

Before running the analysis I defined four narratives as an interpretive lens for each triple $(\hat{\tau}^c, \tau_F^c, \tau_I^c)$: informality amplifies (A), replication of the Danish null (B), the effect concentrated in the informal channel (C), and stratified gains among skilled formal workers (D). Their formal definitions, in signs and magnitudes, are in Section 3.8. The central empirical contribution is the decomposition of the dose by baseline formality in the eight countries; the narratives discipline its reading, and the assignment of a narrative to each country is reported for all eight, labeled as an exploratory reading.

The stylized task-based framework of Section 2 organizes the predictions and connects them to the formality channel in LAC.

The article’s sharpest contribution is its contrast with Humlum and Vestergaard (2025a): the Latin American configuration (high informality, fragmented wage setting, weak social insurance) is the opposite of the Danish one. If the Danish null is universal, it should survive in every LAC country; if instead it reflects Danish institutions rather than LLM capabilities, the eight estimates should diverge in directions predictable from each country’s formality and digital readiness, and should do so at the margin where the institutions differ, which is the price of labor rather than the quantity. The article also provides external validity for the US estimates of Hartley et al. (2026) and Chen, Kane, et al. (2025) in a developing-region context, and takes the step the Latin American literature, stalled at descriptive exposure (Ciaschi et al., 2025; Pizzinelli et al., 2023), had not: from predicted exposure to realized causal effects.⁴

The article makes five contributions. First, it offers the first causal estimates of LLM exposure on labor market outcomes for Latin America, bridging the descriptive exposure literature (Azuara Herrera et al., 2024; Ciaschi et al., 2025; Egaña-delSol & Bravo-Ortega, 2026) and the OECD-centered causal literature (Chen, Kane, et al., 2025; Hartley et al.,

⁴The project’s second article, in preparation, estimates individual-level and distributive effects on the microdata of the same surveys, with heterogeneity by gender, age, education, income, and firm size; the present work concentrates on the aggregate equilibrium at the occupation level.

2026; Humlum & Vestergaard, 2025a). Second, it applies the continuous-treatment DiD of Callaway, Goodman-Bacon, and Sant’Anna (2025) to LLM exposure country by country; the previous causal literature has relied on binary cutoffs and has not exploited the eight-country comparison that the Latin American context allows. Third, it elevates baseline formality from a subgroup check to the central margin of within-country heterogeneity, with an imputation protocol that avoids using a post-treatment outcome as a conditioning variable. Fourth, it builds a crosswalk and a harmonized, reproducible database for eight countries (2021–2025) that future Latin American researchers can extend. Fifth, the four interpretive narratives are defined a priori, disciplining the reading of the empirical patterns against an ex post specification search. **Two layers of credibility cut across these contributions: the exposure measure is validated against observed use of the technology, occupation by occupation and country by country, with Anthropic Economic Index microdata (Fig. 1), and the design is falsified directly by rebuilding the entire cascade over 2015–2019 with a fictitious launch (Section 5.7).**

2 Theoretical Framework

The stylized task-based framework below organizes the empirical predictions and clarifies why the labor market response to LLMs in Latin America is ex ante ambiguous. It adapts Acemoglu and Autor (2011) and Acemoglu and Restrepo (2018, 2022) to a general-purpose technology that primarily reconfigures nonroutine cognitive tasks. The model is deliberately compact: it disciplines the empirical specification rather than delivering a calibration.

2.1 Task-based production

Final output in occupation o combines a continuum of tasks $j \in [0, N]$. Each task is performed by human labor or by capital, which includes traditional automation and, from 2022, software built on LLMs. Occupation o is a task bundle with a fixed vector of shares $\{s_{oj}\}_{j \in [0, N]}$ with $\sum_j s_{oj} = 1$, where s_{oj} is the share of occupation o ’s output generated by task j . Occupational technology is fixed in the short run; labor demand governs reallocation across occupations. **Recent models endogenize task content within the occupation (Freund & Mann, 2026); the fixed-bundle assumption is the appropriate short-run approximation for a three-year window.**

The wage in occupation o is the sum of task-specific compensation:

$$w_o = \sum_j s_{oj} \cdot p_j, \tag{1}$$

where p_j is the equilibrium price of task j . Before LLMs, p_j depends on the productivity of human labor in task j and on local demand for goods that use it intensively. The framework follows Autor et al. (2003) in treating the routine-nonroutine distinction as the axis of previous automation waves, and Acemoglu and Autor (2011) in allowing capital to substitute for labor in any task with a sufficiently productive technology.

2.2 LLMs as a new technology

In November 2022 ChatGPT’s launch makes a subset of tasks $J^{\text{LLM}} \subset [0, N]$ performable by software built on LLMs. The complementary exposure score of Eloundou et al. is the share of occupation o ’s tasks in this subset:

$$\beta_o \equiv \sum_{j \in J^{\text{LLM}}} s_{oj}. \quad (2)$$

The subset is defined by a time criterion: β_o counts tasks in which at least 50 percent of the required time can be saved through software *built on LLMs*, in contrast to direct LLM exposure (α_o) or the upper bound (ζ_o); see Eloundou et al. (2024). [The choice of \$\beta\$ over \$\alpha\$ and \$\zeta\$ as the treatment intensity for LAC is developed in Section 3.2.](#)

The arrival of LLMs has two opposing effects on the wage in exposed occupations. The *displacement effect* ($\tau_d > 0$) operates because firms now produce tasks $j \in J^{\text{LLM}}$ at lower cost using software, reducing labor demand there; it scales with β_o , so occupations with a larger share of automatable tasks face stronger displacement. The *reinstatement effect* ($\tau_r > 0$) operates because firms create new tasks complementary to the technology: prompt design, output verification, supervision, integration with existing workflows. Humans perform these tasks alongside the software, raising labor productivity in exposed occupations.

The net wage response is the difference between the channels, scaled by exposure:

$$\Delta w_o = (\tau_r - \tau_d) \cdot \beta_o. \quad (3)$$

The sign and magnitude of $\tau_r - \tau_d$ are empirical objects, not theoretical predictions ([Agrawal et al., 2019](#)). The framework only pins down the shape across occupations: the response is proportional to β_o in absolute value, so high-exposure occupations move more, in either direction, than low-exposure ones.

2.3 Heterogeneous adoption: the formality channel

Adoption of LLM-augmented software is not free at the firm level: subscription fees, integration with existing systems, worker training, and IT support require fixed-cost investment, so firms with weak credit access, low capitalization, and scarce managerial capacity face prohibitive costs. In Latin America this heterogeneity maps onto the formal-informal divide: formal firms are larger, better capitalized, and more likely to operate in sectors where LLM-augmented software is already commercialized; informal firms are smaller, cash-constrained, and often lack the digital infrastructure these tools require.

The framework therefore predicts asymmetric short-run effects. Within the formal sector both channels operate: $\tau_d^F > 0$ generates partial displacement and $\tau_r^F > 0$ the creation of complementary tasks, with relative magnitudes that may favor tertiary-educated workers able to perform the new supervision tasks. [Usage evidence supports this hierarchy: work-related ChatGPT use is concentrated among users with tertiary education and in well-paid professional occupations \(Chatterji et al., 2025\), a gradient that the Danish adoption survey replicates in its wage dimension \(Humlum & Vestergaard, 2025b\), and matches the profile](#)

of formal employment in the region, so productive adoption enters through the sector where the framework places it. Within the informal sector, direct adoption is close to zero in the short run, so $\tau_d^I \approx 0$ and $\tau_r^I \approx 0$ in the partial-equilibrium sense. Informal workers may nonetheless be affected indirectly. Three spillover channels matter, each with its own effect on informal employment and wages. The first is *market-share capture*: adopting formal firms become more productive and shift demand away from informal producers, reducing informal employment and wages; this is the mechanism of the dual view of informality, in which more productive formal firms displace informal ones as the economy develops (La Porta & Shleifer, 2014), and it also operates in models where both sectors compete within the same industry (Ulyssea, 2018). The second is the *demand and income spillover*: productivity gains make formal goods cheaper and raise formal workers' income, expanding demand for complementary informal goods and services (subcontracting, inputs, household consumption), which raises informal employment and wages. The third is *labor supply reallocation*: displaced formal workers may relocate to the informal sector, a mobility margin widely documented for the region, with intensity varying over the cycle and across episodes (Bosch & Maloney, 2010; Fiess et al., 2010); that supply shift raises informal employment but pushes informal wages down. Cross-sector job search also links wages in the two sectors, so a formal shock reaches the price of informal labor even without massive reallocation (Meghir et al., 2015). The net effect is therefore ambiguous for informal employment (the first channel contracts it, the other two expand it) and for informal wages (the second raises it, the first and third depress it). The decomposition has a testable implication: a rising informal wage in exposed occupations, such as that observed in Brazil, Colombia, and Ecuador in Table 6, is hard to reconcile with dominant labor supply reallocation and points instead to the demand spillover, or to selection among the workers who remain. These three spillovers translate the canonical informality literature to the LLM shock; the review by Ulyssea (2020) and the institutional diagnoses for the region (Levy, 2008; Perry et al., 2007) further ground the formal-informal partition as a structural, not frictional, feature of the Latin American labor market.

The net comparison is:

$$\Delta w_o^F - \Delta w_o^I = \left[(\tau_r^F - \tau_d^F) - (\tau_r^I - \tau_d^I) \right] \cdot \beta_o. \quad (4)$$

Whether the differential is positive or negative depends on the relative strength of direct adoption in formal firms (which can go in either direction) and of the spillovers onto informal firms, whose sign is ambiguous in both employment and wages, depending on the balance of the three channels. The sign is an empirical question in each country. The full map of channels and their signs over both sectors is in Fig. A1 (Section A). The diagram also records a nuance the simplification $\tau_d^I \approx \tau_r^I \approx 0$ omits: direct informal adoption is not exactly zero, since free-access chat interfaces are within reach of own-account workers without fixed-cost investment; the framework assumes this light adoption does not alter the occupation's task structure at the pace of the enterprise software that β measures, so the direct informal channel is second order relative to the spillovers.

2.4 Who bears the adjustment: incumbents and entrants

Equations (1) to (3) carry a single wage per occupation, so they cannot say who inside the occupation absorbs a change in the price of tasks. That is a substantive omission in a segmented labor market, where the wage of an established worker and the wage at which a vacancy is filled are set by different mechanisms: the first is anchored by a contract, by tenure and by the downward rigidity that both impose within a horizon of a few years, while the second is negotiated afresh at each hire and can track the current marginal product of the occupation. The distinction costs little to add and changes what the framework predicts.

Let occupation o employ a mass $1 - \lambda_o$ of incumbents and hire a flow λ_o of entrants each period. Entrants are paid the spot value of the occupation's task bundle, so their wage absorbs the full response $(\tau_r - \tau_d)\beta_o$ of equation (3). Incumbents are paid a protected wage that passes through only a fraction $\phi_o \in [0, 1]$ of that response, with $\phi_o = 0$ under full protection and $\phi_o = 1$ when the contract is renegotiated freely. The occupation's average wage then moves by

$$\Delta \bar{w}_o = [\lambda_o + (1 - \lambda_o)\phi_o](\tau_r - \tau_d)\beta_o, \quad (5)$$

and the composition term λ_o does the work: even with $\phi_o = 0$, so that no established worker loses a cent, the measured average wage of the occupation falls whenever entrants are paid less and the occupation hires. This resolves a tension that equation (3) alone could not accommodate, since with one wage per occupation employment and the wage must move together, while the pairing of stable or rising formal employment with a falling formal average wage is exactly what the data show.

P5 (who bears the adjustment). Where the exposure response is negative, the decline concentrates among entrants, and the earnings of incumbents move by a fraction ϕ_o of it, falling by nothing where contractual protection is complete. The prediction is contestable in a way the earlier ones are not: it fails if incumbents and entrants lose the same amount.

The same parameter yields an institutional comparative static that travels across countries. When the entry wage is rigid, whether by a binding floor or by norms of pay compression, the firm cannot adjust the price at which it hires and adjusts the quantity instead, so the shock surfaces in fewer hires. When the entry wage is flexible and unemployment insurance is weak, so that a worker's outside option is thin, the price of entry gives way first. The framework therefore predicts that the same technological shock appears as a quantity in the first kind of labor market and as a price in the second, which is the comparison [Section 6](#) draws between the United States and the region.

2.5 Testable predictions

The framework delivers four predictions that the empirical specification in [Section 3](#) evaluates.

P1 (across occupations, within country). The labor market response in country c is proportional to β_o :

$$\frac{\partial Y_{ot}^c}{\partial \beta_o} \gtrless 0$$

depending on whether reinstatement dominates displacement, offsets it exactly, or is dominated by it. Here Y_{ot}^c is the labor market outcome of occupation o in quarter t of country c : log employment, average weekly hours, or the log of the average monthly wage, measured at the cell level. It is the dependent variable of the estimating equation in Section 3.3; Section 4.2 details its construction. The sign may differ across countries.

P2 (formal vs. informal, within country). Short-run adoption is concentrated in the formal sector. The framework predicts $|\tau^F| > |\tau^I|$ over the two-year horizon where formal adoption is relevant. In countries where formal adoption is also low, both channels are muted.

P3 (cross-country comparison of the estimates). Countries with higher formality rates and better digital infrastructure should exhibit a larger $|\tau|$ than those with lower formality and weaker infrastructure. This guides the reading of the eight country estimates $\{\hat{\tau}^c\}_{c=1}^8$ presented side by side; their formal test against country-level indicators of formality and digital readiness, which eight units cannot support, is left for future work. Cross-country heterogeneity is a result in its own right, not a robustness check.

P4 (skill heterogeneity within the occupation). Within an exposed occupation, tertiary-educated workers are better positioned to perform the new reinstatement tasks (supervision, prompting, verification) and therefore experience a smaller displacement effect than secondary-educated ones. The margin developed in Section 5 is baseline formality (Section 2.3); the interaction with education that would directly test P4 is not testable with occupational cells and is left for the project’s second paper, on individual microdata.

The framework also reconciles the divergent evidence of the introduction: the within-firm gains from US controlled trials correspond to high- τ_r environments in formal firms, and the Danish null of Humlum and Vestergaard (2025a) to an environment where τ_d and τ_r approximately offset. LAC differs on both margins: adoption is heterogeneous across countries and formality strata, and the variation in informality (from 25 percent in Uruguay to more than 70 percent in parts of the Andean region) generates ex ante predictions that differ by country, so pooling would average over heterogeneity that the theory predicts to be first order: each of the eight country estimates is an independent test of the same mechanism. The macro counterpart of this framework for the region is Bakker et al. (2025): if AI raises formal-sector productivity, it can compress informality in the aggregate; this article estimates the micro-occupational reverse of that hypothesis.

3 Empirical Strategy

The identification strategy is a difference-in-differences (DiD) design with continuous treatment and a single adoption date, estimated *country by country*: not a pooled cross-country panel but eight independent analyses, one per survey. Within each country the data are repeated cross sections (only Mexico’s ENOE is a rotating panel), so the same person is not followed over time. Occupations within a country are exposed to different intensities of LLM capability, measured by Eloundou’s complementary exposure score β , and the launch of ChatGPT on November 30, 2022 is the treatment date common to the eight countries ($t^* = 2022Q4$, coded as pre-treatment because two thirds of it precede the launch; treatment

is effective from 2023Q1). Identification comes from within-occupation variation in how exposure interacts with the post-launch indicator, and the cross-country comparison from the side-by-side presentation of the eight $\hat{\tau}^c$ on three outcomes: employment, hours, and monthly wages, at the occupation-quarter cell level.

3.1 Sample and data

The sample consists of eight labor force surveys harmonized to a common quarterly frequency (2021Q1–2025Q4). The surveys, the country selection criteria, and the temporal and occupational harmonization are documented in [Section 4](#); here I set down the two sample design decisions.

The first is the time window. Periods prior to 2020 are excluded even where data exist: COVID-19 differentially affected high-exposure cognitive occupations (migration to remote work, sectoral reallocation) relative to low-exposure manual and service occupations, so mixing pre-COVID and post-COVID years would compare two structurally distinct equilibria. The 2021Q1 start date draws both the pre- and the post-treatment periods from the post-COVID steady state. The trade-off is a short pre-period: at most [eight quarters between 2021Q1 and 2022Q4, that is, seven pre-treatment coefficients plus the reference quarter in the event study](#) (Colombia, with a panel starting in 2022Q1, has fewer: three pre-treatment coefficients), since all share the rolling-quarter frequency of [Table 1](#). I mitigate this by elevating the Honest-DiD bounds of Rambachan and Roth ([2023](#)) from robustness to primary identification and by exploiting a longer post-period of twelve quarters.

The second is the unit of observation: in country c , the individual worker i at time t , with the 4-digit ISCO-08 occupation $o(i)$ recorded in the survey, aggregated to the (occupation, time) cell for the event-study specifications.

3.2 Treatment variable

The treatment is continuous occupational exposure to LLMs, measured by Eloundou’s complementary exposure score β_o ([Eloundou et al., 2024](#)). The score is the share of an occupation’s tasks for which at least 50 percent of the required time can be reduced through software *built on LLMs* (LLM-augmented applications), distinct from α_o (direct LLM exposure) and ζ_o (the upper bound); see Table 1 of [Eloundou et al. \(2024\)](#). Throughout the paper, $E_o \equiv \beta_o$ is the exposure variable in the estimating equation.

The complementary channel (β) captures the diffusion of LLM-augmented software (third-party productivity tools, customer relationship and accounting platforms, service chatbots, and code assistants built on LLM APIs) through which workers experience LLM capability without necessarily typing into a chat interface. In Latin America, direct ChatGPT use lags OECD levels, but multinational firms deploy LLM-augmented enterprise software, so β is the correct treatment intensity here, and more conservative than ζ . The direct exposure measure α is not appropriate here: it captures only the use of an LLM in a chat interface, whereas the object of interest is the complementary channel, whose penetration in the region

is greater. The score is mapped from O*NET to ISCO-08.⁵ After the crosswalk, 98.2 percent of 4-digit ISCO codes with positive employment in at least one of the six countries with a verified occupational crosswalk match at least one O*NET occupation; unmatched codes receive the average β of their 3-digit group, so all 4-digit codes in that group share the same imputed exposure. This affects the 1.8 percent of codes without a 4-digit match, not the occupations that concentrate employment. The crosswalk extends the Chile-Mexico-Peru pilot validated in Azuara Herrera et al. (2024). Brazil and Bolivia, whose classifiers are national adaptations of ISCO-08 (Section 4.1), use the native code directly with the same 3-digit fallback; effective coverage of 97.9 and 94.1 percent.

The design’s direct empirical precedent is Acemoglu, Autor, et al. (2022), which uses base-line occupational exposure in long differences on AI vacancies; this paper retains that logic with labor survey microdata, the continuous treatment framework of Callaway, Goodman-Bacon, and Sant’Anna (2025), and the country-by-country design. Johnston and Makridis (2026) is the closest comparable in terms of the dose: they apply the same Eloundou exposure score in a DiD on industry-by-state cells in the United States.⁶

The post-treatment indicator is $\text{Post}_t = \mathbb{I}[t \geq 2023\text{Q1}]$. The launch quarter, 2022Q4, is coded as pre-treatment in the primary specification: two thirds of it precede November 30, so coding it as treated would dilute the dose of the post period. The coding errs conservatively in the opposite direction: any effect materializing in December 2022 is absorbed into the pre period and biases τ toward zero. In the event study it is the reference period (Section 3.4); in the synthetic DiD of Section B and its pre-COVID placebo it is excluded entirely as a transition buffer.⁷

3.3 Primary specification: country-by-country continuous DiD

The primary estimator is a two-way fixed effects (TWFE) regression with continuous dose, applied separately to each of the eight countries and interpreted within the identification framework of Callaway, Goodman-Bacon, and Sant’Anna (2025). The treatment date is identical across countries, $t^* = 2022\text{Q4}$, since the launch of ChatGPT in November 2022 was a single global event, and all occupations in a country share that date (single treated cohort). Callaway, Goodman-Bacon, and Sant’Anna (2025) show that, under their *strong* parallel trends assumption (Section 3.5), the coefficient on the interaction $\beta_o \cdot \text{Post}_t$ is a weighted average, with positive weights summing to one, of the average causal responses (ACRT) along the dose distribution; under standard parallel trends, that decomposition additionally incorporates selection bias terms across dose groups. The TWFE weights do not coincide with the dose density among the treated and are sensitive to the mass of occupations with zero exposure, so τ^c must be read as a weighted summary of causal responses, not as

⁵The internal pipeline routes via SOC-2010 as a bridge, but the user-facing crosswalk is O*NET-to-ISCO-08.

⁶Other exposure measures (the patents of Webb (2019), the AIOE of Felten et al. (2021), and the ILO measure of Gmyrek et al. (2023), defined directly on ISCO-08) agree with that of Eloundou et al. (2024) in the coarse ranking of exposed occupations; α and ζ are reported as robustness in Table A3.

⁷Humlum and Vestergaard (2025a) and Hartley et al. (2026) code the launch quarter as treated; the pre coding adopted here is the more conservative of the two.

the average ACRT of the treated population. To bound what that implicit weighting may generate, [Section B](#) estimates the summary that Callaway, Goodman-Bacon, and Sant’Anna (2025) recommend, the $ACRT^{\text{glob}}$ with dose density weights, from a flexible regression of the long difference on the dose among treated occupations: the paper’s eleven significant aggregate estimates retain their sign under that aggregation ([Table A12](#)). The specification estimated within country c is

$$Y_{ot}^c = \tau^c \cdot (E_o \cdot \text{Post}_t) + \alpha_o^c + \alpha_t^c + \varepsilon_{ot}^c, \quad (6)$$

where Y_{ot}^c is one of the three primary outcomes defined in [Section 4.2](#) (log expanded employment, average weekly hours, or the survey-weighted average of individual log monthly wages) in the (occupation, time) cell, $E_o \equiv \beta_o \in [0, 1]$ is the Eloundou complementary exposure score of the 4-digit ISCO occupation, $\text{Post}_t = \mathbb{I}[t \geq 2023\text{Q1}]$, α_o^c and α_t^c are occupation and time fixed effects within the country, and ε_{ot}^c is an idiosyncratic error. Each cell aggregates all employed workers in the occupation and quarter, weighted by the survey expansion weight, with no adjustment for demographic composition: compositional changes are part of the measured effect. For wages, the most relevant composition margin (entrants versus incumbents) is quantified with the individual worker-level test of [Section B](#).

The coefficient τ^c is thus a weighted summary of causal responses. The occupation fixed effect absorbs the level of E_o , time-invariant within an occupation, and the country’s baseline occupational wage structure; the time fixed effect absorbs country-specific macroeconomic shocks, including the post-COVID recovery and inflation. A more demanding fixed effects structure, occupation by quarter (α_{ot}^c), would absorb the treatment itself and is not estimated; the cross-country comparison occurs at the level of the eight estimated values $\hat{\tau}^c$, not within a single regression.

The eight $\hat{\tau}^c$ are presented side by side in [Section 5](#), and the cross-country variation in $\hat{\tau}^c$ organizes the exposition. These aggregates are not, however, the level at which the paper’s main claims rest: the evidentiary hierarchy stated below places them on the estimates aimed directly at the segment’s own outcome, and the margin that orders the reading is the price, the monthly wage, rather than the quantity.

Cells are weighted by survey-expanded employment for population representativeness. Since contemporaneous employment is itself an outcome of the treatment, [Section B](#) re-estimates the aggregate and the channels with fixed baseline weights (each occupation’s average pre-treatment employment); all significant coefficients retain sign and significance. Standard errors are clustered at the level of the 4-digit ISCO occupation within the country, the unit at which E_o varies ([Abadie et al., 2023](#)). Estimation runs on cells rather than on individual microdata because the treatment $E_o \cdot \text{Post}_t$ varies only at the occupation level: employment-weighted cells with occupation-clustered standard errors deliver the same coefficients as an individual-level regression of the corresponding worker-level outcome, such as hours or the log wage, without treating co-workers as independent observations. Expanded employment has no worker-level counterpart, since it is a cell total rather than the mean of an individual variable.

3.4 Secondary estimators

Two secondary estimators triangulate the primary result within each country. The first is the binary split DiD of Callaway and Sant’Anna (2021): [occupations with exposure above the country’s employment-weighted median \(\$p50\$ \)](#),

$$D_o = \mathbb{I}[E_o > \text{median}^c(E)], \quad (7)$$

form the treated group, with common treatment date 2023Q1; those at or below the median play the never-treated role in the estimator. Their exposure is not zero, so the contrast is between high and low dose rather than between treated and untreated units, and its magnitude is not directly comparable to the per-unit coefficient of the continuous specification. The group-time ATT of Callaway and Sant’Anna (2021) is estimated on the employment-weighted occupation panel, aggregated to the overall average ATT, with the estimator’s own [bootstrap intervals](#). It is reported because it is the design of Hartley et al. (2026), Hui et al. (2024), and Liu, Xu, et al. (2025), allowing a direct comparison of magnitudes.⁸ The primary continuous specification is unaffected by this choice.

The second is a TWFE event study with dose-time interaction, estimated within each country. With a single treated cohort, as here where the eight samples share the global date $t^* = 2022Q4$, the cross-cohort contamination problems that motivate Sun and Abraham (2021) do not apply; [for continuous treatments, de Chaisemartin et al. \(2026\) develop related estimators comparing switchers and stayers](#). The specification is:

$$Y_{ot}^c = \sum_{\substack{k=-K \\ k \neq 0}}^K \theta_k^c \cdot (E_o \cdot \mathbb{I}[t - t^* = k]) + \alpha_o^c + \alpha_t^c + \varepsilon_{ot}^c, \quad (8)$$

where $t^* = 2022Q4$ is [the last pre-treatment quarter](#) and θ_0^c is [normalized to zero](#). After the temporal harmonization to rolling quarters described in [Table 1](#), seven of the eight countries share a common event window of $k \in [-7, +12]$ quarters around the launch, and Colombia, whose panel begins in 2022Q1, contributes only $k \in [-3, +12]$, with $k = 0$ as [the reference](#). The expected pattern is flat θ_k^c for $k < 0$ (parallel pre-trends) and a step change at $k = +1$ (2023Q1) that persists or grows.

3.5 Identifying assumptions

The specification identifies τ^c under four assumptions, evaluated separately within each country.

Parallel trends. Within country c , for any exposure level e , the counterfactual trajectory of treated occupations would have evolved in parallel with that of zero-exposure occupations absent the LLM shock. Formally, $\mathbb{E}[Y_{ot}^c(0) - Y_{o,t^*}^c(0) \mid E_o = e] = \mathbb{E}[Y_{ot}^c(0) - Y_{o,t^*}^c(0) \mid E_o = 0]$

⁸I use the country-specific median rather than the theoretical $> 50\%$ threshold because Eloundou’s β has a long left tail: only between 5 and 10 percent of ISCO-08 occupations exceed 0.5, so a $> 50\%$ split produces very unbalanced groups and large standard errors. The median split, used by Liu, Xu, et al. (2025), delivers balanced cells while preserving the binary interpretation.

for all e and all $t > t^*$. Occupations with exactly zero exposure concentrate between 0.5 and 6 percent of employment depending on the country, so in practice the estimator compares dose levels against one another rather than treated units against an untreated group. That comparison across doses requires the *strong* version of parallel trends of Callaway, Goodman-Bacon, and Sant’Anna (2025), under which counterfactual trajectories do not vary systematically with the dose, a more demanding assumption than standard parallel trends relative to the untreated; see also the discussion in Baker et al. (2026). This, together with the short pre-period, is why the Honest-DiD bounds are the primary criterion. The assumption is defended within each country with the pre-trends event study of equation (8), a joint Wald test that $\theta_k^c = 0$ for $k < 0$, and partial identification bounds following Rambachan and Roth (2023); the main claims for country c require a country-specific breakdown value $\bar{M}^c \geq 2$. The breakdown value $\bar{M}^c$ is the largest multiple of the maximum pre-treatment violation that the robust interval tolerates without including zero; requiring $\bar{M}^c \geq 2$ admits post deviations of up to twice the largest deviation observed before treatment, a demanding threshold. Following Roth (2022), pre-trend tests are read with caution: failure to reject the null is not evidence of parallel trends, only that the test lacks power against relevant alternatives. As a direct test of the assumption, Section 5.7 rebuilds the full design on the 2015–2019 vintages of seven of the eight countries and imputes a false shock in 2018Q1, mirroring the actual launch: the paper’s central pattern does not appear in the no-shock world, and the historical violations of parallelism observed there, zero-mean oscillations that do not accumulate bias, provide an empirical anchor for $\bar{M}$ complementing the preregistered threshold (Table 11).

Because no estimate in the article meets the $\bar{M}^c \geq 2$ threshold, that criterion cannot by itself select which results are asserted, and it is worth stating the evidentiary hierarchy that replaces it before any coefficient is read. A first tier carries the article’s claims: results estimated directly on the sector’s own outcome (Section 5.1), surviving the pre-COVID placebo (Section 5.7) and the false discovery rate correction within their family. A second tier is reported and interpreted, but explicitly as the weaker margin: results that hold in the baseline specification yet depend on the functional form imposed on the heterogeneity, or that fail one of those checks. A third tier is exploratory and labeled as such wherever it appears: the joint reading of the five patterns, articulated after observing the Peruvian data, and the assignment of countries to the interpretive narratives, which the a priori definition disciplines but does not validate. The threshold is retained as a reporting standard rather than a selection rule, and every table reports the breakdown value so the reader can apply a stricter criterion than the article’s.

No anticipation. Workers in country c did not adjust labor supply or employment outcomes in anticipation of ChatGPT. The launch was exogenous to LAC labor markets: pre-launch access was restricted to OpenAI employees and selected external testers, none of whom operated in the region at the scale needed to move aggregate outcomes.⁹

Exposure exogeneity. The exposure score β is constructed from O*NET task content (the United States, 2022–2023) using GPT-4 and a human rubric. Neither the task content nor the scoring procedure is a function of country c ’s labor market outcomes in 2023. [What the design](#)

⁹A weak residual anticipation channel via the GPT-3.5 API, released in March 2022, is bounded by the event-study coefficients θ_{-2}^c and θ_{-1}^c within each country.

identifies is therefore the differential post-launch trajectory of more exposed occupations, not the effect of realized LLM adoption: the quarter fixed effects absorb shocks common to all occupations within a country, so any unobserved post-2022 shock whose incidence covaries with β remains a confounder. Task content within ISCO codes is assumed comparable across countries; Azuara Herrera et al. (2024) show for Chile, Mexico, and Peru that exposure scores remain consistent after the crosswalk to national classifiers, without this equating the occupational contexts of the two regions. A risk distinct from exogeneity is measurement error: the dose reaches each occupation through a chain of crosswalks with imputations, and classical error attenuates τ^c toward zero, a conservative direction. Section A probes it by instrumenting $\beta \times \text{post}$ with $\zeta \times \text{post}$; the significant estimates are insensitive to the contrast (IV rows of Table A3). The exercise is robustness to an alternative score rather than a bound on the measurement error: both scores are built on the same task source and travel through the same crosswalk chain, so the error component common to that chain is not purged.

SUTVA. The stable unit treatment value assumption rules out spillovers across occupations within a country. The sign of the resulting bias is not determined in general: if spillovers move the outcomes of less exposed occupations in the same direction as the direct response, the estimated contrast understates the true response, and if they move them in the opposite direction it overstates it. Higher colleague productivity does not by itself fix the sign for employment, wages, or hours. Berg et al. (2018) suggest that as long as the technology performs a small fraction of the economy’s tasks, its general equilibrium effects are modest. SUTVA is the highest-risk assumption and is acknowledged as such for each country-level estimate.

3.6 Within-country heterogeneity: baseline formality as the central margin

Baseline occupational formality is the central dimension of within-country heterogeneity wherever formal-informal segmentation is non-trivial. I use *baseline* rather than contemporaneous formality because the latter is itself an outcome of LLM exposure: a worker may transition from formal to informal because of the treatment, and conditioning on a post-treatment outcome is a bad control.

The operational measure is the baseline occupational formality share: $F_{o,2021}^c$, the survey-weighted share of formal workers in occupation o in country c ’s base year (2021 in seven countries; 2022 in Colombia, whose panel begins that year, still prior to the shock), computed on the cell panel under the harmonized definition of Section 4.3. It is an occupation-level measure, common to the eight surveys and predetermined with respect to the launch. Sensitivity to the definition of formality (restricting it to salaried workers, the best-measured component across surveys) is reported in Section B.

The same equation is also estimated on outcomes built only from workers of a given formality status: within each occupation and quarter, the expanded employment of formal workers, their average log monthly wage and their average weekly hours, and the analogous cells for informal workers. That version replaces the interaction with a direct sector-specific outcome, so its coefficient is the effect on the segment rather than effect modification by the occupation’s baseline formality, at the cost of conditioning on a status measured contempo-

aneously. It is the level at which the evidentiary hierarchy places the article’s main claims, and Section 5.1 reports it.

The interaction specification, estimated separately by country, is

$$Y_{ot}^c = \tau_F^c \cdot (E_o \cdot \text{Post}_t \cdot F_{o,2021}^c) + \tau_I^c \cdot (E_o \cdot \text{Post}_t \cdot (1 - F_{o,2021}^c)) + \alpha_o^c + \alpha_t^c + \varepsilon_{ot}^c, \quad (9)$$

where τ_F^c and τ_I^c are the weighted summaries of causal responses for the baseline formal and informal channels within country c (same reading as τ^c in Section 3.3), and $F_{o,2021}^c$ is as defined above. Since the two regressors sum to $E_o \cdot \text{Post}_t$, the difference $\tau_F^c - \tau_I^c$ is identified from the cross-sectional variation of F_o across occupations at the same dose; its causal reading therefore requires parallel trends also in the formality dimension: that, absent the shock, occupations with different baseline formality but the same exposure would have evolved in parallel. The horse race with the tertiary education share in Section B evaluates the most natural confounder (skill); the sector $\times$ quarter robustness check absorbs the second (sectoral composition); and a second horse race with the teleworkability measure of Dingel and Neiman (2020) evaluates the third (the differential recovery of remote work), with the formality gradient surviving in all eight countries. The four narratives of Section 3.8 are evaluated within each country from the signs and magnitudes of $(\tau^c, \tau_F^c, \tau_I^c)$.

3.7 Robustness layer

The robustness layer rests on triangulation between the primary and the two secondary estimators within each country. I report the Honest-DiD bounds of Rambachan and Roth (2023) with the breakdown value $\bar{M}^c$ for each main country-level result, together with the joint pre-trends test from the event study (equation (8)) and the binary specification of Callaway and Sant’Anna (2021) (equation (7)), whose convergence with the continuous specification indicates that the result does not depend on the functional form imposed on the dose. As a safeguard against the multiple testing implied by 72 main contrasts, Section 5.6 additionally reports which estimates survive the false discovery rate correction of Benjamini and Hochberg (1995) within each family of contrasts.¹⁰

3.8 Four interpretive narratives, evaluated by country

The identification strategy is designed to speak to four empirical patterns, defined a priori and *evaluated within each of the eight countries*: each reports which of the four its $(\hat{\tau}^c, \hat{\tau}_F^c, \hat{\tau}_I^c)$ supports. The empirical contribution is therefore eight independent tests of the same narratives; the assignment of each country to a narrative is an interpretive reading, placed at the exploratory tier of the evidentiary hierarchy just stated, not the paper’s main claim.

Narrative A – Informality amplifies. In country c : $|\hat{\tau}^c| > 2\%$ and $\tau_I^c < \tau_F^c < 0$. Informal workers absorb the shock more than formal workers in country c .

¹⁰All estimation is in R: the continuous estimator uses `fixest`, the Honest-DiD bounds `HonestDiD`, and the event study dose-time interactions in `fixest` (with a single treated cohort the correction of Sun and Abraham (2021) is not required); the binary estimator uses the `did` package. Code and data are distributed in the replication package.

Narrative B – Null replication. In country c : $|\hat{\tau}^c|$ is small and statistically indistinguishable from zero in both channels. Country c is a replication in LAC of the null of Humlum and Vestergaard (2025a).

Narrative C – Null in formal, large in informal. In country c : $\tau_F^c \approx 0$ and τ_I^c significantly negative. Country c identifies a channel that the Danish and US designs cannot detect because their informal sectors are negligible.

Narrative D – Stratified gains. In country c : $\tau^c > 0$ on log wages for high-exposure occupations, concentrated among urban formal workers with tertiary education ($\tau_F^{c, \text{tertiary, urban}} > 0$); employment effects close to zero. LLMs act as a skill-biased productivity enhancer for the protected segment in country c .

Defining them a priori disciplines the reading: which one each country supports is decided by the realized pattern, not by a specification search. If the same narrative obtains in all eight countries, that is evidence of the mechanism’s universality; if different countries support different narratives, of the role of each country’s institutions.

4 Data

This section describes the eight harmonized country-level microdata sets and the occupational harmonization on which the identification strategy of Section 3 rests. The unit of harmonization is the four-digit ISCO-08 occupation, the ILO classification that bridges the region’s heterogeneous national classifications and the LLM exposure measures built on O*NET and SOC. For each country I build a panel of cells (native four-digit occupation, quarter) over 2021Q1–2025Q4, with the country-specific trims detailed in Table 2 and with the complementary exposure β from Eloundou et al. (2024) assigned to each native occupational classification through the common mapping to four-digit ISCO-08. For Peru, the main result of Section 5 comes from the Encuesta Nacional de Hogares (ENAH) of the Instituto Nacional de Estadística e Informática (INEI), with national coverage.

4.1 The eight harmonized surveys

Two criteria selected the eight countries. First, occupational coding compatibility: five code occupations in a national classification based on ISCO-08 and are brought to the four-digit ISCO-08 standard through each country’s official correspondence tables: Costa Rica (Encuesta Continua de Empleo, ECE), Colombia (Gran Encuesta Integrada de Hogares, GEIH), Ecuador (Encuesta Nacional de Empleo, Desempleo y Subempleo, ENEMDU), and Uruguay (Encuesta Continua de Hogares, ECH), plus Peru (ENAH), whose CNO-2015 mapping is detailed below. Mexico (Encuesta Nacional de Ocupación y Empleo, ENOE) is the exception: it publishes in its own national classification, the SINCO (Sistema Nacional de Clasificación de Ocupaciones), which I map to four-digit ISCO-08 with the official SINCO–ISCO crosswalk of the Instituto Nacional de Estadística y Geografía (INEGI); the coverage of that chain (employed workers with an assigned exposure β) is 92.5 percent in the 2021–2025 ENOE, and codes without a match are assigned at the three-digit level (Section 3.2). Brazil (PNAD Contínua, IBGE) and Bolivia (Encuesta Continua de Empleo, INE) code in national adapta-

tions of ISCO-08: the COD (Classificação de Ocupações para Pesquisas Domiciliares) takes ISCO-08 as its reference by design (IBGE, 2025) and the COB 2009 is the Bolivian adaptation of ISCO-08 (INE, 2009). The correspondence is also verified empirically: 432 of the 434 base groups in the official COD structure match an ISCO-08 code digit by digit, and the two remaining national-adaptation codes account for 0.3 percent of PNADC employment; in Bolivia, the ECE field coding (the five-digit COB code truncated to its first four digits) resolves as a three-digit ISCO-08 minor group for 86 percent of employment and as a four-digit unit group for 13 percent, exactly the levels served by the hierarchical exposure fallback of Section 3.2. Treating both codes as ISCO-08 therefore follows the classification’s institutional design, not a conjecture. The difference relative to the six countries with their own correspondence tables is one of granularity, not validity: the Bolivian dose inherits from its field coding a predominantly three-digit resolution, a caveat kept explicit throughout. Other LAC countries use national classifications without published crosswalks to ISCO (Chile’s CIUO-CL) or publish at coarser detail. Second, representativeness: the eight surveys are nationally representative.

The Peruvian data come from the ENAHO’s employment and income module 500, with occupation coded in the four-digit CNO-2015 (variable `p505r4`), stable over the entire window, and mapped to ISCO-08 via INEI’s Anexo 3.

The eight labor force survey databases are harmonized to a common quarterly frequency, with quarters defined by the standard rolling calendar: Q1 = January–February–March, Q2 = April–May–June, Q3 = July–August–September, Q4 = October–November–December. Temporal harmonization follows a single rule: quarterly surveys are used as is; in monthly surveys the three months of each calendar quarter are aggregated into a single cell; annual surveys with continuous fieldwork are split into four quarters by interview month. The “native frequency” in Table 1 is the periodicity at which each institute publishes its survey (monthly, quarterly, or annual), and the “rolling quarter” is not a sliding window but the fixed calendar quarter just defined. Table 1 summarizes the native surveys and the rule applied to each country, over a common sample window from 2021Q1 to 2025Q4.¹¹

4.2 Harmonized outcome variables

The design estimates the treatment effect on three labor outcomes, all measured at the cell level (four-digit ISCO-08 occupation, quarter) on each survey’s employed population. Each cell aggregates employed workers with a valid occupation and an assigned exposure β ; all cell aggregates below are weighted by the survey expansion weight.

Employment is the log of the cell’s expanded employment, the sum of the survey expansion weights of the employed workers who compose it; it measures the estimated employed population the occupation represents in the quarter. *Hours* are the average of hours usually worked per week in the main occupation among the cell’s employed workers. The *monthly wage* is the average of the log of monthly labor income from the main occupation; it is not

¹¹One caveat: the ENAHO and ECH expansion weights are calibrated to the year (or month), so the Peruvian and Uruguayan quarterly aggregates are representative by construction rather than by sampling design. This is second-order for identification, since quarter fixed effects absorb the level, but it raises per-quarter sampling variance; validation against the official INEI and INE series is left for future work.

Table 1: Labor force surveys and temporal harmonization

Country	Survey	Native frequency	Harmonization to rolling quarter
Bolivia	ECE	Quarterly (panel)	As is
Brazil	PNADC	Quarterly	As is
Colombia	GEIH	Monthly	Aggregate 3 months → 1 quarter
Costa Rica	ECE	Quarterly	As is
Ecuador	ENEMDU	Monthly (continuous)	Aggregate 3 months → 1 quarter
Mexico	ENOE	Quarterly	As is
Peru	ENAH0	Annual (continuous fieldwork)	Split by interview month → 4 quarters
Uruguay	ECH	Annual (continuous fieldwork)	Split by interview month → 4 quarters

Notes: Quarters defined as Q1 = January-February-March, Q2 = April-May-June, Q3 = July-August-September, and Q4 = October-November-December. The sample window covers 2021Q1 to 2025Q4. [The launch quarter, 2022Q4, is coded as pre-treatment in the primary specification \(two thirds of the quarter pre-date the launch; Section 3.2\), serves as the reference period in the event study, and is excluded in the synthetic DiD.](#) For Uruguay and Peru the native survey (ECH and ENAH0) is annual but fieldwork is spread over the year, which allows each household to be assigned to a quarter by its interview month; for Colombia and Ecuador the three calendar months of each quarter are aggregated weighting by each month’s survey expansion weight (the sampling weight that expands each observation to the population it represents); Costa Rica and Mexico already publish quarterly.

Table 2: The eight harmonized surveys and their mapping to four-digit ISCO-08

Country	Survey	Classification (β coverage)	Active ISCO-08 4d	Mean N per quarter
Bolivia	ECE	COB (94.1%)	128	20,766
Brazil	PNADC	COD (99.2%)	399	199,118
Colombia	GEIH	ISCO-08 (99.2%)	388	80,175
Costa Rica	ECE	ISCO-08 (96.6%)	183	7,580
Ecuador	ENEMDU	ISCO-08 (99.5%)	330	41,031
Mexico	ENOE	SINCO (92.5%)	376	173,551
Peru	ENAH0	CNO-2015 (99.4%)	192	14,141
Uruguay	ECH	ISCO-08 (99.0%)	235	6,395

Notes: [The native frequency of each survey and its harmonization rule to the fixed calendar quarter are reported in Table 1.](#) The native classification column gives the source occupational system and, in parentheses, the crosswalk’s *coverage*: the percentage of employed workers with an exposure score β assigned through the correspondence chain (native classification → four-digit ISCO-08 → β). The four surveys based on ISCO-08 use each country’s official correspondence tables; Mexico (SINCO via INEGI) and Peru (CNO-2015 via INEI’s Anexo 3) use crosswalks between distinct classifications, with unmatched codes assigned at the three-digit level. [The Brazilian COD and the Bolivian COB 2009 are national adaptations of ISCO-08, treated as ISCO-08 at the code level with hierarchical fallback to three digits \(Section 4.1\).](#) Coverage exceeds 96% in six of the eight countries and is 92.5% in Mexico and 94.1% in Bolivia; in Colombia it is computed over the 2022Q1–2025Q4 window, since the 2021 database uses a two-digit classification without correspondence and is left out of the panel. “Active ISCO-08 4d” is the number of four-digit occupations with positive employment in the cell panel. “Mean N per quarter” is the average number of sample observations (unexpanded). The sample window is 2021Q1–2025Q4 for Costa Rica, Ecuador, Mexico, Peru, Uruguay, Bolivia, and Brazil, and shortens to 2022Q1–2025Q4 in Colombia given microdata availability. Only cells meeting the minimum observation threshold are retained.

divided by hours, so the outcome captures monthly pay, not the hourly price. For Peru, where the ENAHO reports labor income annually, monthly income divides by twelve the sum of income earned as a dependent and as an independent worker.

All three outcomes are conditional on being employed, so selection through entry into and exit from employment is not directly identifiable with repeated cross-sections. Wages are restricted to positive values and hours to plausible ranges, discarding each survey’s non-informative codes; no percentile winsorization is applied. Only cells with at least ten sample observations are retained, to prevent occupations with very little mass from dominating the weighted estimators’ variance.

4.3 Harmonized definition of formality

The central margin of within-country heterogeneity is baseline formality status (Section 3.6). To keep the formal-informal partition comparable across surveys with different instruments, I apply a harmonized rule based on contributory social security coverage: an employed person is *formal* if she is a wage earner contributing to a pension system, or an independent worker, self-employed, or employer whose business is registered with the tax authority; everything else is informal. The rule is common to the eight countries; only the native variables that implement it differ. In Colombia (GEIH), Costa Rica (ECE), Ecuador (ENEMDU), and Peru (national ENAHO), formality combines the wage earner’s pension contribution with the independent worker’s tax registration, exactly as above. In Uruguay (ECH) a single variable, contribution to the retirement fund, covers wage earners and independent workers. In Mexico (ENOE) I adopt INEGI’s official classification of formal employment, based on access to social security through employment. In Bolivia (ECE), as in Uruguay, affiliation to the pension administrator (Gestora de pensiones, AFP) is the single official criterion for wage earners and independent workers. In Brazil (PNADC) the IBGE’s official position-in-occupation variable (VD4009, which distinguishes employment with and without a signed carteira de trabalho) combines position and registration for wage earners, exactly as above; for employers and the self-employed I use the contribution to a social security institute (instituto de previdência, VD4012), analogous to the other countries’ tax registration. Formality is measured at the 2021 baseline and, for the seven repeated cross-sections, is imputed at the cell level as the *share* of formal workers in 2021 within the worker’s demographic-occupational cell (Section 3.6).

Table 3 quantifies the margin on the paper’s own panels. Employment-weighted baseline informality ranges from 23.4 percent in Uruguay to 73.2 percent in Ecuador, and in all eight countries it is lower in high-exposure than in low-exposure occupations (10.2 versus 36.1 percent in Uruguay; 42.2 versus 70.4 in Mexico; 63.8 versus 82.4 in Ecuador; 24.5 versus 52.3 in Brazil; 53.9 versus 82.2 in Bolivia). This gradient, predetermined with respect to the launch of ChatGPT, is the structural fact motivating the formality decomposition as the central contribution: variation that the designs behind the existing causal estimates cannot exploit, since informal employment is marginal in the high-income economies where they were produced.

Table 3: Baseline informality and gradient by exposure, own data

Country	Base year	Informality (%)	High exposure	Low exposure	Gap (p.p.)
Bolivia	2021	68.1	53.9	82.2	28.3
Brazil	2021	38.8	24.5	52.3	27.8
Colombia	2022	62.2	48.2	75.9	27.7
Costa Rica	2021	44.5	37.1	51.7	14.6
Ecuador	2021	73.2	63.8	82.4	18.5
Mexico	2021	56.5	42.2	70.4	28.2
Peru	2021	71.6	61.3	81.7	20.5
Uruguay	2021	23.4	10.2	36.1	25.9

Notes: Informal employment shares weighted by expanded employment from this article’s own cell panels, in each country’s base year, under the harmonized pension-based definition of formality (Section 4.3). High (low) exposure: occupations above (below) the country’s employment-weighted median of β . In all eight countries informality is systematically lower in the more exposed occupations.

4.4 Validating the exposure against observed AI usage

The design requires separating the potential exposure that β measures from each country’s realized adoption. Exposure is a technological property of the occupation’s task content, common across countries by construction; that invariance guarantees its exogeneity. Realized adoption, by contrast, depends on digital infrastructure and the fabric of firms, and scales the reduced form: the same β can produce different $\hat{\tau}^c$ because the effective dose differs. For within-country identification this is innocuous, since quarter fixed effects absorb the national adoption level; for the cross-country comparison it is the context that organizes prediction P3. That this dose is already material in the region has direct support: with OpenAI’s internal data, Chatterji et al. (2025) document that ChatGPT reached about 10 percent of the world’s adult population by mid-2025, growing disproportionately fast in middle-income countries like those in this sample. Table A1 brings that context to the country level, combining digital infrastructure with two measures of realized adoption: the Claude per-capita usage index from the Anthropic Economic Index (Anthropic, 2026) and the worldwide per-capita message ranking for ChatGPT. The ordering is informative for P3: Uruguay and Costa Rica, with the most marked formal-channel responses (the Uruguayan reversal and the Costa Rican formal wage decline), lead in both infrastructure and adoption, while Ecuador and Bolivia, the secular drift and the weakest P pattern, have the lowest adoption (0.47 and 0.53), with Mexico, the null case, barely above them (0.61). Brazil is the partial exception, with a formal-channel response as sharp as those of Costa Rica and Peru despite intermediate adoption (0.91), which suggests that the absolute size of the exposed formal labor market, not only per-capita adoption intensity, mediates the aggregate effect’s magnitude. Finally, Fig. 1 validates the exposure measure against observed usage: an occupation’s share of Claude usage, relative to its share of employment, increases with its β both in the aggregate (rank correlation of 0.30 over the 643 occupation-country pairs) and within seven of the eight countries (correlations from 0.25 to 0.36, including Brazil, $\rho = 0.29$), so the occupations the rubric declares exposed are those that in fact use the technology. One

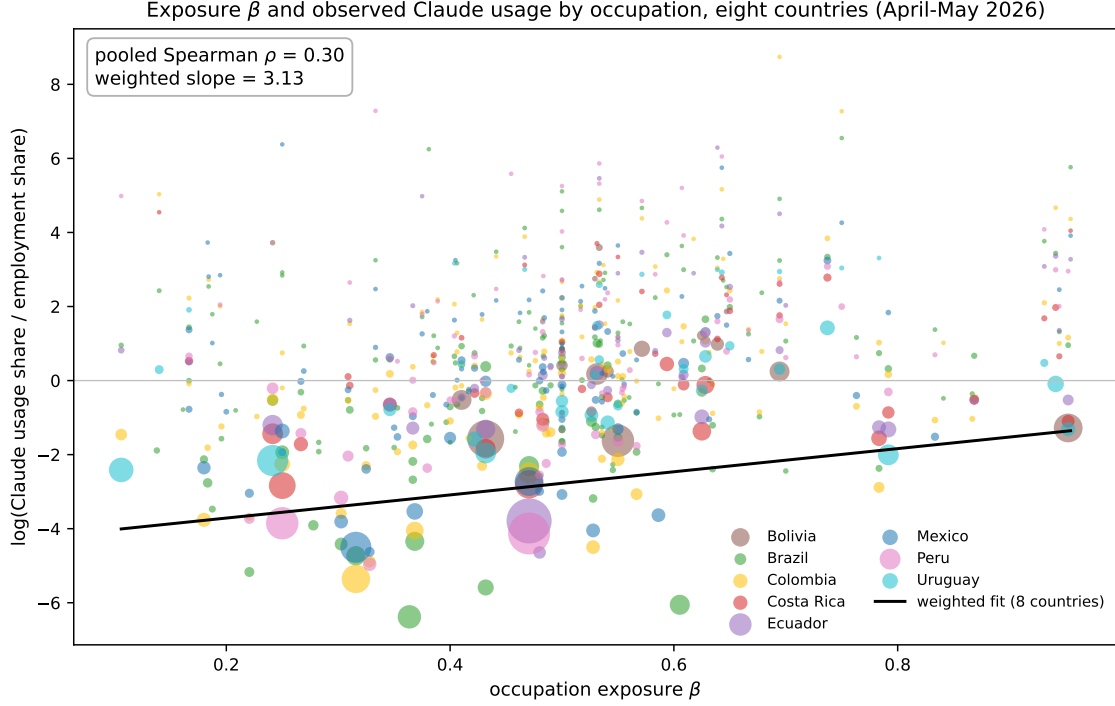


Figure 1: *Validation of the exposure β against observed Claude usage by occupation, eight countries.* Each point is a detailed SOC occupation in a country (643 occupation-country pairs); the vertical axis is the log ratio of the occupation’s share of the country’s Claude usage (Anthropic Economic Index, April-May 2026 average, conversation-to-occupation classification from Handa et al. (2025)) to its share of the country’s employment (own harmonized databases, 2025). Point size is proportional to the within-country employment share, color to the country; the black line is the pooled weighted fit and ρ the Spearman rank correlation over the 643 pairs. The slope is insensitive to weighting (3.13 employment-weighted, 3.30 unweighted; ρ identical by construction), so the gradient is not driven by large occupations. Occupations with usage suppressed for privacy in the AEI (small cells) are left out, so the matched sample covers between 34 and 58 percent of each country’s usage, except in Bolivia (11 percent, the caveat in the text).

limitation of this validation should be stated: the AEI usage is measured in April and May of 2026, after the outcome window closes, so it corroborates the ranking of exposure against observed use but not the timing of adoption within the estimation period. The exception is Bolivia ($\rho = -0.11$), where only 14 SOC occupations are matched to published usage in the AEI (10.7 percent of the country’s usage, versus 34–58 percent in the rest of the sample), so the Bolivian correlation lacks power and is reported as a data limitation, not as evidence against the exposure measure.

The same cross-tabulation yields a fact connecting adoption to the paper’s thesis: recorded usage of generative AI is concentrated in formal employment. Comparing, by country, the formality of employment in occupations with Claude usage above the AEI’s publication threshold with that of total employment (figure in the replication package) shows this. The gap is positive in all eight countries and widest where informality is highest: 44 versus 30 percent in Peru, 41 versus 25 in Ecuador, 51 versus 39 in Colombia, and most extremely

63 versus 32 percent in Bolivia, the most informal country in the sample and the widest gap of the eight; Brazil, the least informal after Uruguay, has instead the narrowest gap (71 versus 64 percent). The structural reading is the theoretical framework’s: productive adoption enters the labor market through the formal segment, so baseline formality is not merely a margin of heterogeneity but the entry gate of the shock itself.

5 Results

This section presents the paper’s main result: country-by-country continuous-treatment estimates for the eight countries (Colombia, Costa Rica, Ecuador, Mexico, Peru, Uruguay, Brazil, and Bolivia), presented first in their direct estimation on cells built separately for formal and informal workers, which is the main evidence, and then in the aggregate and in the decomposition by baseline formality that approach the same question indirectly. What holds across the sample is a price: exposure lowers the formal monthly wage in six of the eight countries, and in five of them the decline survives a placebo built on the same cells, while the employment response is weaker and varies in sign across countries. [Brazil and Bolivia share the same cell panel and specification; their classifiers are national adaptations of ISCO-08 \(Section 4.1\).](#) Each estimation runs on its own harmonized panel of cells (native four-digit occupation, quarter), with the complementary exposure β of Eloundou et al. (2024) assigned through the mapping to four-digit ISCO-08 of [Section 4](#). The regression is TWFE with occupation and quarter fixed effects, weighted by expanded employment and clustered at the occupation level; for Peru I use the quarterly national ENAHO, and formality follows the harmonized pension-based criterion.

5.1 Direct sector-specific estimates

The article’s substantive claim is about formal and informal workers, so the estimates that carry it must be built on their own outcomes. This subsection does that, and it comes first because the evidentiary hierarchy of [Section 3](#) places the main claims here. The two designs that follow approach the same question indirectly: the aggregate of [Section 5.2](#) averages both segments, and the interaction of [Section 5.3](#) keeps the occupation’s total outcome on the left-hand side and modulates the dose by baseline formality, so what it identifies is effect modification by the occupation’s formality rather than the effect on each group of workers. Using the same harmonization, the same formality rule and the same minimum cell size, the occupation-quarter cells are rebuilt *separately* for formal and informal workers, so that the dependent variable becomes the sector’s own employment, wage and hours, and the same continuous equation is estimated on each. [Table 4](#) reports the result.

The main result is that *the fall in the exposed formal wage is the article’s most uniform regularity and survives direct estimation*. The formal sector’s monthly wage falls significantly at the 5 percent level in six of the eight countries (Bolivia -0.070 , Brazil -0.047 , Costa Rica -0.109 , Ecuador -0.232 , Mexico -0.059 and Peru -0.082), with Uruguay a precise zero (-0.001) and Colombia the single exception in sign ($+0.123$). The signs match those of

Table 4: Directly estimated sector-specific effects: formal and informal outcomes separately

Country	Sector	Employment (log)	Monthly wage (log)	Weekly hours
Bolivia	Formal	−0.056 (0.114)	− 0.070 ** (0.031)	−1.080 (0.754)
	Informal	+0.065 (0.165)	−0.004 (0.049)	−1.977 (1.568)
Brazil	Formal	+0.019 (0.038)	− 0.047 ** (0.021)	−0.012 (0.175)
	Informal	+ 0.213 *** (0.071)	+ 0.067 ** (0.029)	+0.676* (0.357)
Colombia	Formal	−0.068 (0.069)	+ 0.123 *** (0.038)	− 0.696 *** (0.221)
	Informal	−0.072 (0.155)	+ 0.148 *** (0.045)	−0.740 (0.629)
Costa Rica	Formal	+0.207* (0.112)	− 0.109 ** (0.046)	−1.557 (1.015)
	Informal	+ 0.266 ** (0.121)	−0.063 (0.137)	−6.384* (3.355)
Ecuador	Formal	+0.147 (0.139)	− 0.232 *** (0.038)	−1.262* (0.727)
	Informal	− 0.286 ** (0.132)	+0.091* (0.050)	−0.629 (1.300)
Mexico	Formal	+0.007 (0.048)	− 0.059 *** (0.017)	+0.527 (0.362)
	Informal	+0.068 (0.068)	−0.018 (0.031)	−0.167 (0.674)
Peru	Formal	+ 0.152 ** (0.077)	− 0.082 ** (0.032)	−1.272 (0.983)
	Informal	+0.177 (0.156)	−0.060 (0.079)	+1.048 (0.880)
Uruguay	Formal	− 0.213 *** (0.072)	−0.001 (0.022)	−0.079 (0.299)
	Informal	+0.102 (0.182)	−0.118 (0.076)	+3.313* (1.790)

Notes: Each coefficient comes from a separate regression of $Y_{ot}^s = \tau^{c,s}(\beta_o \times \text{post}_t) + \alpha_o + \alpha_t$ on occupation-quarter cells built *only* from workers in sector s : that group's expanded employment, its average log monthly wage and its average weekly hours. The dependent variable here is the sector's own outcome, unlike the interaction design of Table 6, where the outcome is the occupation total and baseline formality modulates the dose. Same harmonization, same formality rule and same minimum cell size as the main panel. Occupation-clustered standard errors in parentheses. Worker formality is measured contemporaneously, so these estimates incorporate recomposition between sectors. Coefficients significant at 5 percent in bold. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

the interaction decomposition in the seven countries where it was informative, with smaller magnitudes, so the wage channel does not depend on the functional form imposed on the heterogeneity. Mexico appeared at first to illustrate the value of the exercise, since its aggregate null coexisted with channels indistinguishable from zero while the sector-specific estimate returns a significant formal wage decline. The falsification below withdraws that reading: the Mexican coefficient is the one case in the sample that the placebo contaminates, so the count the article claims is five, not six.

The employment margin is weaker than the interaction suggested. The expansion of formal employment holds in Peru (+0.152, $p < 0.05$) and marginally in Costa Rica (+0.207, $p < 0.10$), while in Brazil, where the interaction returned +0.167 ($p < 0.01$), the direct estimate is an imprecise zero (+0.019, n.s.), and in Bolivia, where the interaction returned +0.188, the direct coefficient flips sign and remains imprecise (−0.056, n.s.). The Uruguayan reversal, by contrast, is confirmed and strengthened (−0.213, $p < 0.01$). Three significant employment responses appear in the informal sector: expansion in Brazil (+0.213, $p < 0.01$) and Costa Rica (+0.266, $p < 0.05$) and contraction in Ecuador (−0.286, $p < 0.05$), the last being the direct counterpart of the Ecuadorian informal channel the aggregate reading already emphasized. The positive informal wage spillover in Brazil (+0.067, $p < 0.05$) and Colombia (+0.148, $p < 0.01$) also survives, with Ecuador marginal (+0.091, $p < 0.10$). The Brazilian one, however, does not clear the sector-level falsification of [Section 5.7](#).

The two estimates are complements and neither dominates. The sector-specific one measures directly what the article wants to claim, but conditions on the worker’s contemporaneous formality, which is post-treatment and can respond to the shock, so it incorporates recomposition between sectors. The baseline-formality interaction is immune to that compositional change, since its regressor is fixed in 2021, but identifies only effect modification. That both agree on the sign of the formal wage channel in seven of eight countries, and on the Uruguayan reversal and the Ecuadorian informal channel, is why the article reads those results as robust; that they differ on the magnitude of formal employment is why the employment expansion is reported as the more fragile margin.

5.2 Aggregate estimates

Why the aggregate hides the result is the natural next question. [Table 5](#) reports the aggregate coefficient $\hat{\tau}^c$ of the interaction $\beta_o \times \text{post}_t$ for each country and primary outcome, pooling formal and informal workers in a single cell.

Aggregate effects are heterogeneous across countries and, in several cases, small or statistically indistinguishable from zero. Employment is positive and significant in Costa Rica (+0.302), Peru (+0.283) and Brazil (+0.123, $p < 0.01$), negative and significant in Ecuador (−0.164) and Uruguay (−0.163), and not distinguishable from zero in Colombia, Mexico and Bolivia (+0.069, n.s.). The monthly wage falls significantly in Costa Rica and Peru, and with less precision in Bolivia (−0.059, $p < 0.10$), and rises in Colombia. These aggregates conceal, in several countries, systematic differences between the formal and informal channels.

Since the treatment is continuous, $\hat{\tau}^c$ is the effect of one unit of β ; a more interpretable contrast is the within-country interquartile jump, from the 25th to the 75th percentile of the employment-weighted distribution of β (between 0.19 and 0.35 points of β depending on

Table 5: Aggregate estimates $\hat{\tau}^c$ (continuous treatment $\beta \times \text{post}$), TWFE DiD by country

	log employment	log monthly wage	weekly hours	N cells	N ISCO
Bolivia	0.069 (0.121)	-0.059* (0.030)	-1.067 (1.153)	2,198	128
Brazil	0.123*** (0.042)	-0.025 (0.025)	0.135 (0.173)	7,458	399
Colombia	-0.017 (0.089)	0.118*** (0.033)	-1.468*** (0.356)	5,400	388
Costa Rica	0.302*** (0.098)	-0.178*** (0.050)	-4.027*** (1.130)	2,540	183
Ecuador	-0.164** (0.082)	-0.058 (0.041)	-0.611 (0.741)	5,514	330
Mexico	0.064 (0.049)	-0.017 (0.024)	0.325 (0.478)	7,021	376
Peru	0.283** (0.112)	-0.097** (0.049)	0.135 (0.675)	2,707	192
Uruguay	-0.163** (0.073)	0.015 (0.027)	0.319 (0.451)	2,650	235

Notes: Each $\hat{\tau}^c$ is the coefficient on the interaction $\beta_o \times \text{post}_t$ in a TWFE regression with occupation and quarter fixed effects, weighted by expanded employment and with standard errors clustered at the occupation level (in parentheses, below each coefficient). The unit of treatment variation is each country’s native four-digit occupation code, to which the exposure β of Eloundou et al. (2024) is assigned via mapping to ISCO-08. Post = 2023 onward (ChatGPT launch, Nov. 2022). Peru corresponds to the national ENAHO reconstructed at quarterly frequency. “N cells” is the number of (occupation, quarter) observations in the regression and “N ISCO” the number of four-digit occupations that identify it. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

the country).¹² On this metric, employment in the most exposed occupations relative to the least exposed moves by roughly +7.9% in Costa Rica, +6.0% in Peru and +3.8% in Brazil, and about -3.0% in Ecuador and -5.5% in Uruguay. For the monthly wage the same jump implies around -2.0% in Peru, -4.4% in Costa Rica and -1.7% in Bolivia, and +3.4% in Colombia, while weekly hours fall by about 0.4 hours in Colombia. These are moderate but nontrivial magnitudes for a two-year horizon, and it is their sign, not only their size, that varies across countries.

Weekly hours, the third outcome in Table 5, deserve their own reading. The aggregate effect is negative and significant only in Colombia (-1.468, $p < 0.01$) and Costa Rica (-4.027, $p < 0.01$); in the other six countries it is small and indistinguishable from zero, including Brazil (+0.135, n.s.) and Bolivia (-1.067, n.s.). In the interquartile contrast that is about 0.4 fewer weekly hours in Colombia and 1.0 in Costa Rica, most versus least exposed occupations. Where the intensive margin responds, it does so in the direction the exposure metric itself predicts: β measures the share of tasks whose execution time can be cut in half, so a shorter workweek in exposed occupations is the direct manifestation of the time savings. The absence of an hours effect elsewhere matches the Danish evidence, which likewise finds no intensive-

¹²The jump is the difference between the 75th and 25th percentiles of occupation-level β , weighted by pre-treatment expanded employment of the cell panel; it ranges from 0.19 (Ecuador) to 0.35 (Uruguay). Log coefficients are converted to percentages as $\exp(\hat{\tau} \cdot \text{IQR}) - 1$.

Table 6: Heterogeneity by baseline formality: formal and informal channels by country

		log employment	log monthly wage	weekly hours	N (cells/ISCO)
Bolivia	$\hat{\tau}_F$ (formal)	0.188 (0.116)	− 0.092 ** (0.039)	0.180 (0.901)	2,198 / 128
	$\hat{\tau}_I$ (informal)	−0.152 (0.142)	0.003 (0.055)	− 3.393 * (1.801)	
Brazil	$\hat{\tau}_F$ (formal)	0.167 *** (0.057)	− 0.069 *** (0.024)	0.080 (0.227)	7,458 / 399
	$\hat{\tau}_I$ (informal)	−0.100 (0.118)	0.202 *** (0.069)	0.421 (0.876)	
Colombia	$\hat{\tau}_F$ (formal)	0.111 (0.096)	0.125 *** (0.045)	− 2.714 *** (0.323)	5,400 / 388
	$\hat{\tau}_I$ (informal)	−0.313 (0.249)	0.101 ** (0.046)	1.401 ** (0.685)	
Costa Rica	$\hat{\tau}_F$ (formal)	0.399 *** (0.103)	− 0.267 *** (0.060)	− 4.089 *** (1.209)	2,540 / 183
	$\hat{\tau}_I$ (informal)	−0.082 (0.155)	0.205 (0.133)	−4.185 (2.732)	
Ecuador	$\hat{\tau}_F$ (formal)	−0.060 (0.084)	− 0.208 *** (0.043)	−0.628 (0.580)	5,514 / 330
	$\hat{\tau}_I$ (informal)	− 0.341 *** (0.114)	0.193 *** (0.058)	−0.557 (1.646)	
Mexico	$\hat{\tau}_F$ (formal)	0.106 * (0.057)	−0.007 (0.026)	0.880 *** (0.338)	7,021 / 376
	$\hat{\tau}_I$ (informal)	−0.045 (0.072)	−0.044 (0.038)	−1.136 (1.188)	
Peru	$\hat{\tau}_F$ (formal)	0.406 *** (0.089)	− 0.144 *** (0.033)	−0.619 (0.801)	2,707 / 192
	$\hat{\tau}_I$ (informal)	0.003 (0.168)	0.009 (0.098)	1.636 (1.118)	
Uruguay	$\hat{\tau}_F$ (formal)	− 0.179 *** (0.065)	0.021 (0.024)	0.139 (0.399)	2,650 / 235
	$\hat{\tau}_I$ (informal)	0.216 (0.188)	−0.130 (0.145)	4.393 * (2.386)	

Notes: Coefficients from the specification that decomposes the dose $\beta_o \times \text{post}_t$ into a formal channel ($\times F_o$) and an informal one ($\times (1 - F_o)$), where F_o is the employment-weighted occupational formality share in the base year (first observed year of each country). Formality follows the harmonized pension-based definition: a wage earner who contributes to a pension system, or a self-employed worker/employer whose business is registered with the tax authority. Each cell reports the coefficient with its standard error in parentheses to the right. Occupation and quarter fixed effects, employment weighting, and standard errors clustered at the occupation level. “N (cells / ISCO)” reports the (occupation, quarter) observations and the four-digit occupations of each country’s regression; the panel is the same as in Table 5. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

margin adjustment. This reading must be taken with the caution imposed by Table 8: the pre-trends for hours are intermediate between employment’s (cleaner) and the wage’s (more contaminated).

5.3 Heterogeneity by baseline formality

The third design corroborates the first from a different angle, and carries its own limit. Table 6 decomposes the exposure dose into a formal channel ($\beta_o \times \text{post}_t \times F_o$) and an informal one ($\beta_o \times \text{post}_t \times (1 - F_o)$), where F_o is the employment-weighted formality share of the occupation in each country’s base year.

The interaction decomposition points above all to the *formal channel*, where exposure to LLMs tends to reduce the monthly wage and, less uniformly, to expand employment in occupations with predominantly formal baselines. Of the two movements, only the wage one holds when the outcome is measured on formal workers directly, as the sector-specific estimation of Section 5.1 shows; the employment one weakens there and is the least firm reading of this table. Formal employment is positive in Costa Rica (+0.399, $p < 0.01$), Peru

(+0.406, $p < 0.01$), Brazil (+0.167, $p < 0.01$), Mexico (+0.106, $p < 0.10$), Colombia (+0.111, n.s.) and Bolivia (+0.188, n.s.); the formal wage is negative in Peru (-0.144 , $p < 0.01$), Costa Rica (-0.267 , $p < 0.01$), Ecuador (-0.208 , $p < 0.01$), Brazil (-0.069 , $p < 0.01$) and Bolivia (-0.092 , $p < 0.05$). The informal channel is, in this decomposition, generally weaker and less systematic in sign, with reversals (a positive informal wage in Colombia and Ecuador, negative informal employment in Ecuador, and most markedly the Brazilian informal wage, +0.202, $p < 0.01$, the largest informal-channel reversal in the entire sample).

The Brazilian informal spillover deserves the main qualification: the individual-level composition test of Section B shows it reflects selective retention of informal incumbents, who gain wages with exposure (+0.119, $p < 0.01$) while entrants do not (-0.117 , n.s.), not a generalized improvement of the sector; the sector-level falsification of Section 5.7 adds a more demanding caution, since it contaminates that channel in all three outcomes. The same test makes the formal wage decline (-0.069 , $p < 0.01$) entirely an entry phenomenon: incumbents' wages do not move (+0.026, n.s.) and entrants come in much lower (-0.356 , $p < 0.01$), the most intense entry compression in the sample; Brazil thus replicates, with a different survey, the composition mechanism of pattern P. Its formal employment demands a cautious reading: the pre-trends are strongly rejected and the coefficient attenuates under sector $\times$ quarter, although the synthetic DiD of Section 5.6 confirms it with a clean placebo (Section 5.7).

The cancellation-between-channels reading applies unevenly. In Colombia the aggregate employment null coexists with channels of opposite sign (formal +0.111, informal -0.313), but neither is individually significant, so the cancellation is only suggestive and the null should be read first of all as imprecision; the one Colombian margin where cancellation is demonstrable is hours, where the two channels are significant and of opposite sign. In Mexico there is no cancellation to invoke: both channels are small and indistinguishable from zero in employment and the wage (only formal employment reaches marginal significance at +0.106, $p < 0.10$), so the Mexican null is interpreted as a genuine absence of effect, consistent with Narrative B. Bolivia is a mixed case: formal employment is positive and of the order of pattern P (+0.188) but does not reach conventional significance, most likely because of the small Bolivian sample (2,198 cells over 128 occupations, the smallest of the eight); the informal channel is negative and equally imprecise (-0.152). As in Colombia, there is no arithmetic cancellation, only imprecision in both channels.

The two sign-reversed exceptions in the formal employment channel, Ecuador and Uruguay, are discussed with the interpretive narratives in Section 5.4. The pre-COVID placebo of Section 5.7 distinguishes them: the contraction of exposed Ecuadorian employment already existed in 2015–2019 and should be read as secular drift, while the Uruguayan one does not appear under a false shock and is a phenomenon of the post-launch period. The joint reading of the eight countries yields the paper's central conclusion: segmentation makes the LLM shock operate in materially different ways on the formal and informal sectors, and that differential, not the aggregate, is the contribution.

The formal channel is economically relevant where significant. Under the same interquartile jump in β from Section 5.2, formal employment expands by about +8.8% in the most exposed formal-dominant occupations relative to the least exposed in Peru (0.406×0.207), +10.6% in Costa Rica (0.399×0.252) and +5.1% in Brazil (0.167×0.299), with downward

pressure on the formal wage of -2.9% , -6.5% and -2.0% respectively. These magnitudes belong to the interaction design and should be read as its upper bound: the specification omits the $F_o \times \text{post}_t$ term, and once it is included the formal employment coefficient is absorbed in Costa Rica, Peru and Brazil. The formal wage decline is absorbed with it in Costa Rica and Peru, where the coefficient changes sign and loses significance, and in Bolivia; it survives only in Brazil and Ecuador (Table A13). That is why the direct sector-specific estimates of Section 5.1 are the paper’s main evidence, and the employment magnitudes in this paragraph are reported as a reference point for the interaction design.

The decomposition by formality also reorders hours. The shorter workweek is concentrated in the formal channel: -2.714 ($p < 0.01$) in Colombia and -4.089 ($p < 0.01$) in Costa Rica, while Mexico shows the only positive and significant formal coefficient ($+0.880$, $p < 0.01$). Colombia is also the only country where the two hours channels are significant and of opposite sign (formal -2.714 versus informal $+1.401$, $p < 0.05$), so the Colombian hours aggregate (-1.468) does average opposing channels. Read with the Colombian monthly wage, which rises in both channels, the cut in formal hours implies that hourly pay in the formal channel grows by more than the reported monthly wage. Elsewhere the hours channels are imprecise, in line with the weak intensive margin in the aggregate, with two exceptions: the Bolivian informal hours channel is negative and significant (-3.393 , $p < 0.10$), one of only two country-channel hours combinations reaching significance outside Colombia, Costa Rica and Mexico (the other is the Uruguayan informal channel, $+4.393$, $p < 0.10$), although built on a reduced base of Bolivian informal cells and without a significant formal counterpart ($+0.180$, n.s.).

5.4 Mapping to the interpretive narratives

The central empirical contribution is not any individual $\hat{\tau}^c$ but the cross-country pattern: which of the four interpretive narratives of Section 3.8, or the fifth pattern P defined below, each country supports, judged by its formal and informal channels. Pattern P is the signature none of the four narratives anticipates: employment expansion with downward wage pressure in exposed formal-dominant occupations and a weak or silent informal channel. Unlike the four, defined a priori, pattern P was articulated after observing the Peruvian estimates and is therefore explicitly labeled exploratory. Table 7 summarizes the classification.

Five readings emerge, and the first is the one the rest of the article rests on. *The formal wage channel is the regularity that holds across the sample:* exposure depresses the formal monthly wage in six of the eight countries once the outcome is measured on formal workers directly, and in five of them the decline survives the placebo (Sections 5.1 and 5.7), and the interaction decomposition agrees in sign wherever it is informative. *The canonical signature of pattern P, which pairs that wage decline with a formal employment expansion,* is narrower: Peru, Costa Rica and Brazil display it in the interaction design, in three different economies with three different surveys, but its employment half survives direct estimation only in Peru and, marginally, in Costa Rica. Brazil adds a nuance absent in Peru and Costa Rica, a significantly positive informal wage channel, which the composition test corroborates as selective retention of incumbent informal workers rather than a generalized wage improvement of the sector. Second, *Ecuador supports Narrative C:* the formal employment channel

Table 7: Mapping of each country to the interpretive narratives, based on the channel signature $(\hat{\tau}_F, \hat{\tau}_I)$

Country	Employment (log)		Monthly wage (log)		Supported pattern
	$\hat{\tau}_F$	$\hat{\tau}_I$	$\hat{\tau}_F$	$\hat{\tau}_I$	
Bolivia	0.188	-0.152	-0.092**	0.003	Weak P (only the formal wage channel is significant)
Brazil	0.167***	-0.100	-0.069***	0.202***	P (formal expands, wage falls; positive informal spillover)
Colombia	0.111	-0.313	0.125***	0.101**	D (wage gains)
Costa Rica	0.399***	-0.082	-0.267***	0.205	P (formal expands, wage falls)
Ecuador	-0.060	-0.341***	-0.208***	0.193***	C (formal null, informal contracts)
Mexico	0.106*	-0.045	-0.007	-0.044	B (replication of the null)
Peru	0.406***	0.003	-0.144***	0.009	P (exploratory)
Uruguay	-0.179***	0.216	0.021	-0.130	Inverted P (formal contracts)

Notes: Qualitative classification of each country into the four interpretive narratives (Section 3.8) and pattern P (Section 5.4), based on the signs and significance of the formal and informal channels in Table 6. **Narrative B:** null effects in both channels. **Narrative C:** $\hat{\tau}_F^{\text{employment}} \approx 0$ with a significantly negative $\hat{\tau}_I^{\text{employment}}$. **Narrative D:** positive wage gains with employment close to zero. **Pattern P:** formal employment expands while the formal wage falls, with the informal channel silent. **Inverted P:** formal employment contracts and the informal channel absorbs it. **Weak P:** the sign of formal employment is consistent with pattern P but does not reach conventional significance, while the formal wage does fall significantly. Coefficients in bold significant at least at 5%. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Peru is labeled exploratory because pattern P was articulated after observing its estimates, whereas the four remaining narratives were specified from the theoretical framework prior to the empirical analysis. Brazil additionally exhibits a positive and significant informal wage spillover, absent in Costa Rica and Peru; Bolivia is the country with the smallest number of cells in the sample, which calls for additional caution about the precision of its estimates.

is indistinguishable from zero while the informal one contracts markedly and significantly ($\hat{\tau}_I^{\text{employment}} = -0.341$, $p < 0.01$), precisely the channel Danish and US designs cannot detect because their informal sectors are negligible. The pre-COVID placebo, however, separates the two Ecuadorian pieces: the negative aggregate continues a trend that predates the shock (and disappears under synthetic DiD, which absorbs it by construction), while the informal contraction does not appear in the no-shock world (-0.633 , n.s.), so it is the latter, not the aggregate, that survives the falsification (Section 5.7). Third, *Mexico replicates the null* (Narrative B): effects are small and indistinguishable from zero in both channels, consistent with still-incipient adoption. Colombia comes closest to Narrative D on the margins this design observes, with wage gains and employment close to zero. The match is partial, and the gap should be stated: D requires the gains to be concentrated among urban formal workers with tertiary education, and the sector-specific estimates show them in both segments (formal $+0.123$ and informal $+0.148$, both $p < 0.01$), while the urban and educational dimensions of the narrative are not identified at the occupation-cell level and belong to the individual-level analysis of the companion paper. The Colombian assignment therefore rests only on the combination of wage gains with null employment, not on the stratification that gives D its name.

Fourth, *Uruguay reverses pattern P*: formal employment contracts ($\hat{\tau}_F^{\text{employment}} = -0.179$, $p < 0.01$) while the informal channel is positive but imprecise ($\hat{\tau}_I^{\text{employment}} = +0.216$, n.s.). The reversal is consistent with the framework of [Section 2](#): in the most formal and most digitally prepared economy in the sample, displacement can dominate reinstatement precisely where adoption is most intense, pushing workers toward informality. That reallocation margin is documented for the region, where the informal sector has absorbed employment after formal contractions ([Bosch & Maloney, 2010](#); [Fiess et al., 2010](#)), and the Uruguayan combination of formal contraction with a positive though imprecise informal channel is consistent with that margin operating partially, dampened by the demand spillover. Fifth, *Bolivia supports a weak version of pattern P*: the formal wage falls significantly ($\hat{\tau}_F^{\text{wage}} = -0.092$, $p < 0.05$) and stably under the leave-one-out and the sector-quarter control (where it even strengthens), while formal employment has the right sign (+0.188) but does not reach conventional significance, again for sample size. The informal channel is generally silent, except in hours, where it contracts significantly. Two cautions, discussed in [Section 5.6](#), qualify this reading: the collinearity between formality and tertiary education is among the highest in the sample ($\text{corr}(F_o, T_o) = 0.71$, just below Peru’s), which makes the two gradients hard to separate, and the synthetic DiD yields a significant placebo effect on employment in the pre-COVID exercise, a warning about the estimator’s precision in such a small panel that advises against overinterpreting the sign of its real employment estimate.

What the eight countries share is a price, not a quantity: the exposed formal monthly wage falls in six of them, while the employment response is slight and varies in sign, which is why they do not converge to a single narrative. The response to LLMs in LAC is thus mediated by each country’s formality structure, and that heterogeneity in the quantity margin is of first order, not a second-order nuisance to be averaged away. Pattern P is labeled exploratory because it was articulated after observing the Peruvian data; the four narratives, defined a priori, are evaluated in all eight countries.

5.5 The framework’s predictions against the data

Before turning to robustness, I revisit the four predictions of [Section 2.5](#) in light of the eight countries. **P1** (the response scales with β_o and its sign is an empirical matter) is the prediction the design turns into an instrument, and the data sustain it weakly: in six of the eight countries the dose interaction is significant in at least one outcome, and the sign varies exactly as P1 allowed. **P2** ($|\tau^F| > |\tau^I|$ where formal adoption is relevant) holds in Costa Rica, Peru, Mexico, Uruguay and Bolivia, but is *violated* in the paper’s two most informative cells: Ecuadorian employment (informal -0.341 versus formal -0.060) and the Brazilian wage (informal $+0.202$ versus formal -0.069). The violation does not contradict the framework, whose three spillovers toward the informal sector ([Section 2.3](#)) allow large indirect channels, but it does contradict the hierarchy of magnitudes P2 conjectured: in economies with large informal sectors, the spillover can exceed the direct effect, precisely what OECD designs cannot see. **P3** (greater formality and digital readiness, greater $|\tau|$) receives partial support: Uruguay and Costa Rica, with the highest realized adoption, show the most marked formal-channel responses, and Ecuador and Bolivia, with the lowest, the weakest; Brazil is the partial exception discussed in [Section 4](#), with a sharp response despite

intermediate adoption. A formal test of P3 against country-level indicators would require more units than eight and is left for future work. **P4** (heterogeneity by education within the occupation) is not testable with occupational cells; the horse race of Section B evaluates the educational margin *across* occupations (formality survives the control for skill), and the within-occupation contrast is left for the second paper, on individual microdata. **P5** (who bears the adjustment) is supported, and it is the only prediction the design tests directly on individual microdata. The entrant-incumbent differential is negative and significant in all five countries with tenure data, as the prediction requires. Its free parameter behaves as the framework reads it: pass-through to incumbents is nil in Peru and Brazil ($\phi \approx 0$, incumbent earnings unmoved), partial in Costa Rica ($\phi \approx 0.4$) and in Ecuador ($\phi \approx 0.5$), so contractual protection is not uniform across the region. The prediction would have failed had incumbents and entrants lost the same amount, and they do not in any country. Colombia is the anomaly, with incumbent earnings rising while entrants gain nothing. On balance, the framework emerges strengthened in its architecture (occupational dose, formal-informal partition, spillovers, and the two-tier wage of Section 2.4) and corrected in a conjecture about magnitudes, and that correction is itself the regional finding.

5.6 Parallel trends, Honest-DiD, and robustness of the aggregate effect

Before privileging the decomposition by channels, I assess the credibility of the aggregate variation with the same battery of diagnostics in the eight countries: a continuous-treatment event study (the interaction $\beta_o \times \text{time}$, reference period 2022Q4), the Wald test of parallel trends on pre-treatment coefficients, and the sensitivity bounds of Rambachan and Roth (2023). Colombia, which lacks 2021, contributes three pre-treatment quarters and the rest seven. Table 8 reports the Wald p -value and the Honest-DiD breakdown value $\overline{M}$ for the average post-treatment effect of each country and outcome.

Three regularities emerge. First, *employment* has the cleanest pre-trends: the test does not reject parallelism at 5% in four of the eight countries (Colombia, Ecuador, Mexico and Uruguay) and rejects only marginally in Costa Rica ($p = 0.065$), Peru ($p = 0.067$) and Bolivia ($p = 0.054$), while Brazil is the only country where employment clearly rejects parallelism ($p < 0.001$). Second, the *monthly wage* is the least credible: the test rejects at 5% or less in seven of the eight countries (Colombia, Costa Rica, Mexico, Peru, Uruguay, Brazil and Bolivia) and at 10% in Ecuador, so pre-trends contaminate the aggregate wage reading in the entire sample. Hours are intermediate, with Brazil rejecting at 10% ($p = 0.096$) and Bolivia not rejecting ($p = 0.676$). Third, and decisively, no country or outcome survives the Honest-DiD criterion: the breakdown value $\overline{M}$ never exceeds 0.5 and is zero in most cases, far below the threshold $\overline{M} \geq 2$. It suffices to allow deviations from parallelism comparable to a small fraction of the observed pre-treatment variation for the robust interval of the aggregate effect to include zero. Fig. 2 plots the eight employment event studies, whose pre-treatment dynamics are consistent with this reading.

The binary estimator of Callaway and Sant’Anna (2021) (high baseline β occupations against low-exposure occupations as a never-treated control) delivers the same signal. In Table A2, 22 of the 24 country $\times$ outcome contrasts have bootstrap confidence intervals in-

Table 8: Parallel trends and Honest-DiD robustness by country

Country		log employment	log monthly wage	weekly hours
Bolivia (7 pre periods)	Pre-trends Wald (p)	0.054*	0.000***	0.676
	Honest-DiD breakdown $\overline{M}$	0.0	0.0	0.0
Brazil (7 pre periods)	Pre-trends Wald (p)	0.000***	0.003***	0.096*
	Honest-DiD breakdown $\overline{M}$	0.5	0.0	0.0
Colombia (3 pre periods)	Pre-trends Wald (p)	0.105	0.000***	0.091*
	Honest-DiD breakdown $\overline{M}$	0.0	0.0	0.0
Costa Rica (7 pre periods)	Pre-trends Wald (p)	0.065*	0.029**	0.298
	Honest-DiD breakdown $\overline{M}$	0.0	0.0	0.5
Ecuador (7 pre periods)	Pre-trends Wald (p)	0.139	0.082*	0.333
	Honest-DiD breakdown $\overline{M}$	0.0	0.0	0.0
Mexico (7 pre periods)	Pre-trends Wald (p)	0.404	0.006***	0.009***
	Honest-DiD breakdown $\overline{M}$	0.5	0.0	0.5
Peru (7 pre periods)	Pre-trends Wald (p)	0.067*	0.000***	0.129
	Honest-DiD breakdown $\overline{M}$	0.5	0.0	0.0
Uruguay (7 pre periods)	Pre-trends Wald (p)	0.087*	0.003***	0.000***
	Honest-DiD breakdown $\overline{M}$	0.0	0.0	0.0

Notes: The Wald row reports the p -value of the joint test that all pre-treatment coefficients of the continuous event study ($\beta_o \times \text{time}$, reference 2022Q4) are zero; a high p is consistent with parallel trends. The Honest-DiD row reports the breakdown value $\overline{M}$ of Rambachan and Roth (2023) under relative-magnitude restrictions for the average post effect: the largest $\overline{M}$ for which the robust interval still excludes zero. Cases that survive the $\overline{M} \geq 2$ criterion in bold. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

cluding zero; the exceptions are weekly hours in Costa Rica and employment in Brazil (0.056, CI [0.010, 0.101], consistent with the significance of Brazilian aggregate employment above). The three estimators agree that the *aggregate effect* of LLM exposure is fragile: heterogeneous in sign under continuous TWFE, not robust to minimal deviations from parallelism, and indistinguishable from zero almost everywhere in the binary contrast. Far from weakening the paper’s thesis, this fragility reinforces it: the systematic signal is in the decomposition by formality of Section 5.3, not in the mean effect.

The results may also depend on the exposure measure. Table A3 re-estimates the aggregate $\hat{\tau}^c$ replacing β with the alternative measures of Eloundou et al. (2024), the direct LLM exposure α and the upper bound ζ , holding the specification fixed. The central qualitative finding is stable: the sign of the employment effect is preserved under ζ wherever β was significant (positive in Costa Rica, Peru and Brazil, negative in Ecuador and Uruguay), and the downward wage pressure in Costa Rica, Peru and Bolivia reappears under ζ . α is the noisiest measure, since direct interaction with chat interfaces is still modest in the region; the paper therefore privileges β as the dominant diffusion channel (Section 2.2). The same family of measures also probes the measurement error in the dose: since β reaches each occupation through a chain of crosswalks with imputations, classical measurement error would

Event study of (log) employment by country: $\beta_o \times \text{time}$ interaction, ref. 2022Q4

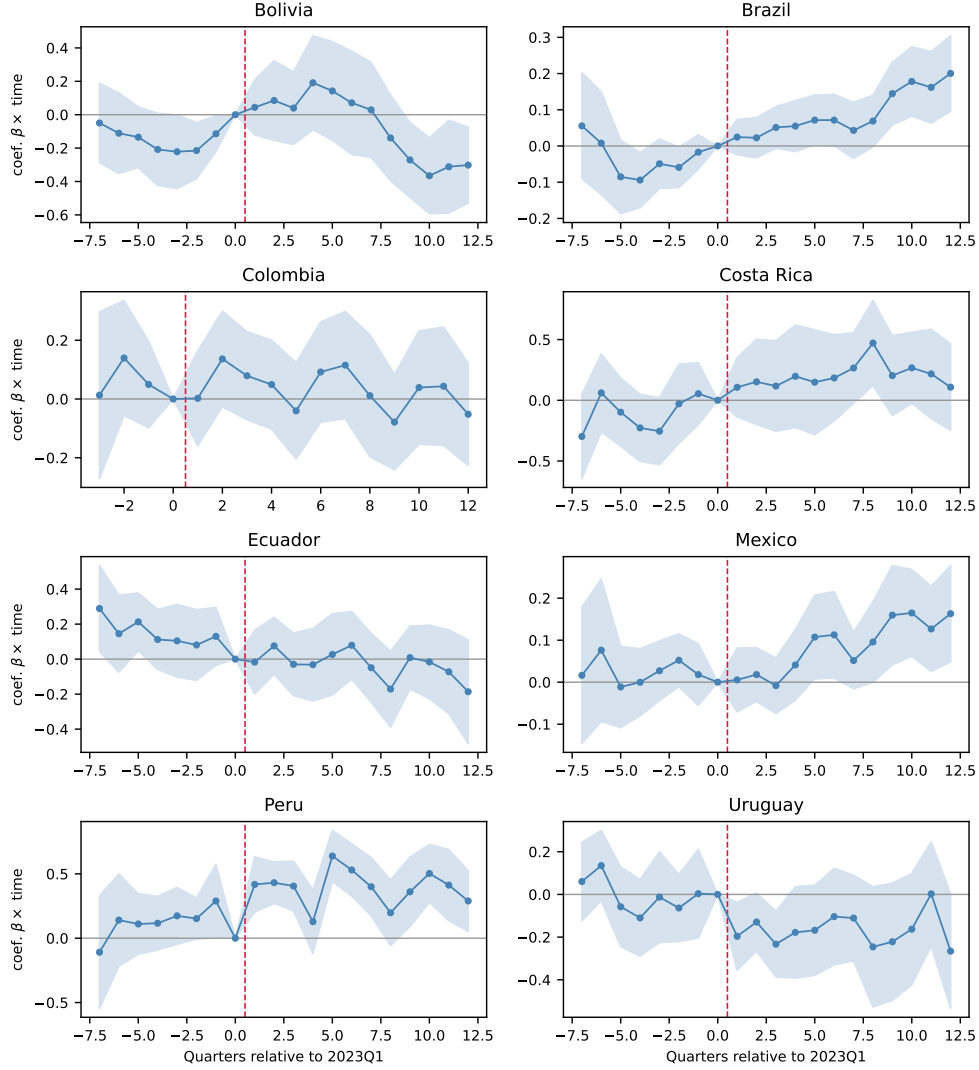


Figure 2: *Event study of (log) employment by country.* Coefficients of the interaction $\beta_o \times \text{time}$ from the continuous-treatment event study on each harmonized panel of cells (native four-digit occupation, quarter), weighted by expanded employment, standard errors clustered at the occupation level. Reference period 2022Q4 (last pre-shock quarter, coefficient set to zero); the vertical red line marks the first post-shock quarter (2023Q1). Bands are 95 percent confidence intervals. Colombia, which lacks 2021, has fewer pre-treatment quarters.

attenuate $\hat{\tau}^c$ toward zero. The IV rows of Table A3 instrument $\beta_o \times \text{post}_t$ with $\zeta_o \times \text{post}_t$, purging the error specific to the rubric weighting (first-stage F between 492 and 1,708): the eleven significant estimates keep their sign and their significance, with a single change of level (Ecuadorian employment moves from 5 to 10 percent), and the magnitudes move within ± 35 percent with no systematic amplification, so there is no sign of first-order attenuation. The alternative measures of Webb (2019), based on tasks and patents, and Felten et al. (2021),

based on skills, require external occupational mappings unavailable here and are left for future work.

Since the systematic signal resides in the decomposition by formality, the same battery of parallel trends and Honest-DiD must be applied to the channel coefficients $\hat{\tau}_F$ and $\hat{\tau}_I$ separately, and not only to the mean effect. Table 9 reports, for each country and outcome, the Wald p -value and the breakdown value $\bar{M}$ of the formal and informal channels, estimated jointly in the same event study that interacts relative time with $\beta_o \times F_o$ and $\beta_o \times (1 - F_o)$.

Three readings follow, and they qualify the strength of the central finding. First, the pre-trends of the *formal employment channel*, where the canonical signature of pattern P is concentrated, are mixed: the Wald test does not reject parallelism at 5% in four of the eight countries (Colombia, Costa Rica, Ecuador and Mexico), with larger p -values than the aggregate's, but rejects in Peru, Uruguay, Brazil and Bolivia, so the formal channel demands additional caution in three of the four countries that support some version of pattern P (Peru, Brazil and Bolivia), with the exception of Costa Rica. Second, the *monthly wage* inherits the aggregate's fragility in both channels: the test rejects parallelism at 5% in almost all country $\times$ channel cells, the sole exception being the Mexican informal channel. Third, the Honest-DiD fragility carries over in full to the channel level ($\bar{M}$ never exceeds 0.5): the channel estimates are interpreted as suggestive, and the central contribution remains the cross-country pattern of Section 5.4, not the point magnitude of any single coefficient.

One cross-cutting safeguard remains. With eight countries, three outcomes and two channels, the paper reports 72 main contrasts, several of which can be expected to reach nominal significance by chance. I therefore apply the false discovery rate correction of Benjamini and Hochberg (1995) within each family. In the family of 24 aggregate estimates, seven survive at the 5 percent level (employment in Costa Rica, Peru and Brazil; the wage in Costa Rica and Colombia; hours in both) and Uruguayan employment joins at the 10 percent level. In the family of 48 channel coefficients, fifteen survive at the 5 percent level, among them all the cells the paper's reading rests on: formal employment and the formal wage in the three pattern P countries (Costa Rica, Peru and Brazil), the Brazilian informal wage spillover, the two Ecuadorian informal channels of Narrative C, and the Uruguayan reversal of formal employment; the Bolivian formal wage survives at the 10 percent level. The cells that do not pass are exactly those the text treats as marginal (Mexican formal employment, informal hours in Bolivia and Uruguay). The a priori narratives, which fix which patterns are sought before estimating, are a design safeguard complementary to this correction.

Section B completes the battery with ten additional exercises: the post-COVID rebound correction (without 2021 and with sector $\times$ quarter fixed effects), formality redefined over salaried workers only, the leave-one-out by major occupational group, the two horse races of the formality margin (against tertiary education and against the teleworkability of Dingel and Neiman (2020)), the individual-level wage composition test, the synthetic DiD of Arkhangelsky et al. (2021), the direct estimation of the $ACRT^{\text{glob}}$ of Callaway, Goodman-Bacon, and Sant'Anna (2025), robustness to seasonality, reweighting by baseline employment, and the contrast with the lower-order interaction $F_o \times \text{post}_t$. The joint verdict is uniform: the decomposition by formality survives every exercise where it was significant, and the country-specific nuances (the sectoral attenuation of Brazilian formal employment, the Bolivian strengthening under controls, the Ecuadorian fragility in the leave-one-out) are

Table 9: Parallel trends and Honest-DiD robustness by channel (formal and informal) and country

Country	Channel		log employment	log monthly wage	weekly hours
Bolivia (7 pre per.)	Formal	Pre-trends Wald (p)	0.047**	0.000***	0.537
		Honest-DiD $\overline{M}$	0.0	0.0	0.0
	Informal	Pre-trends Wald (p)	0.079*	0.000***	0.025**
		Honest-DiD $\overline{M}$	0.0	0.5	0.5
Brazil (7 pre per.)	Formal	Pre-trends Wald (p)	0.000***	0.000***	0.033**
		Honest-DiD $\overline{M}$	0.5	0.0	0.5
	Informal	Pre-trends Wald (p)	0.029**	0.000***	0.061*
		Honest-DiD $\overline{M}$	0.0	0.0	0.0
Colombia (3 pre per.)	Formal	Pre-trends Wald (p)	0.251	0.000***	0.028**
		Honest-DiD $\overline{M}$	0.0	0.0	0.5
	Informal	Pre-trends Wald (p)	0.711	0.002***	0.013**
		Honest-DiD $\overline{M}$	0.0	0.0	0.5
Costa Rica (7 pre per.)	Formal	Pre-trends Wald (p)	0.084*	0.032**	0.640
		Honest-DiD $\overline{M}$	0.5	0.0	0.0
	Informal	Pre-trends Wald (p)	0.041**	0.000***	0.087*
		Honest-DiD $\overline{M}$	0.0	0.0	0.5
Ecuador (7 pre per.)	Formal	Pre-trends Wald (p)	0.549	0.000***	0.851
		Honest-DiD $\overline{M}$	0.0	0.0	0.0
	Informal	Pre-trends Wald (p)	0.055*	0.001***	0.091*
		Honest-DiD $\overline{M}$	0.0	0.0	0.0
Mexico (7 pre per.)	Formal	Pre-trends Wald (p)	0.656	0.000***	0.005***
		Honest-DiD $\overline{M}$	0.5	0.5	0.5
	Informal	Pre-trends Wald (p)	0.000***	0.243	0.020**
		Honest-DiD $\overline{M}$	0.0	0.0	0.0
Peru (7 pre per.)	Formal	Pre-trends Wald (p)	0.000***	0.000***	0.004***
		Honest-DiD $\overline{M}$	0.5	0.0	0.0
	Informal	Pre-trends Wald (p)	0.643	0.001***	0.765
		Honest-DiD $\overline{M}$	0.0	0.0	0.0
Uruguay (7 pre per.)	Formal	Pre-trends Wald (p)	0.007***	0.000***	0.000***
		Honest-DiD $\overline{M}$	0.5	0.0	0.0
	Informal	Pre-trends Wald (p)	0.002***	0.005***	0.350
		Honest-DiD $\overline{M}$	0.5	0.0	0.0

Notes: Repeats the diagnostic of Table 8 on the formal and informal *channels*, estimated jointly in a single continuous-treatment event study that interacts relative time with $\beta_o \times F_o$ (formal channel) and with $\beta_o \times (1 - F_o)$ (informal channel), where F_o is the baseline occupational formality share. The Wald row reports the p -value of the joint test that the channel's pre-treatment coefficients are zero; a high p is consistent with parallel trends. The Honest-DiD row reports the breakdown value $\overline{M}$ of Rambachan and Roth (2023) for the channel's average post effect. Cases that survive the $\overline{M} \geq 2$ criterion in bold. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

discussed there. Section 5.7 then subjects the design to the most demanding falsification in the battery. The binary-estimator and alternative-exposure tables, and the wage and hours event-study figures, are in Section A.

A final question concerns the cross-country reading, which is the article's central contribution: do the estimates genuinely differ, or is their dispersion consistent with a common effect plus sampling noise? Since each country is estimated on a separate survey, the eight estimates are independent and admit Cochran's Q test. Table A14 reports it for the nine

blocks of results. Eight of the nine reject homogeneity, six of them at the one per thousand level, with I^2 between 54 and 91 percent: cross-country dispersion comfortably exceeds estimation uncertainty. The rejection is strongest precisely where the article rests its reading, the formal channel ($Q = 44.4$ for employment, 63.1 for wages and 81.9 for hours, all with $p < 0.001$). The single exception is the informal employment channel ($Q = 9.1$, $p = 0.247$), consistent with its being the weakest and noisiest channel in the battery, and one more reason not to rest conclusions on it.

5.7 Falsification: a pre-COVID placebo

This subsection subjects the design to a direct falsification. I reconstruct the whole cascade (same harmonized surveys, same occupational mapping and dose β , same definition of formality, same cell rules) over the 2015Q1–2019Q4 window, twenty quarters like the real panel, and impute a fictitious launch of generative AI in November 2017, mirroring ChatGPT in November 2022: a false post period from 2018Q1, twelve pre quarters and eight post. In that shock-free world I run the same battery: continuous-treatment TWFE, the joint Wald test on the event-study pre coefficients, and synthetic DiD on the binary split. If the real-period results reflected secular trends of the exposed occupations rather than the shock, the same machinery should find them in 2015–2019. As an *in-time* falsification, the false date is shifted within the pre-period (2017Q1 and 2017Q3): a genuine effect of a specific date should not survive the shift, while a secular drift appears at any date. Colombia is left out: the GEIH prior to the 2021 redesign codes occupation in the two-digit CNO-70, insufficient to assign the dose, the same reason its real panel starts in 2022. Table 10 summarizes the three tests by country and outcome.

Three findings organize the reading. First, and central: *pattern P survives the falsification in Costa Rica and Peru and on both of its margins*. In Costa Rica the false employment effect is an exact zero (-0.003 , $p = 0.98$) against the real $+0.302$, and the false wage effect is *positive* ($+0.124$, n.s.), opposite in sign to the real one (-0.178): the Costa Rican secular drift, if anything, pushes exposed wages upward, so the real estimate would be conservative. In Peru the false employment effect is eleven times smaller than the real one ($+0.025$ against $+0.283$), and the false SDID is zero in both countries. Second, *the placebo differentiates the two sign exceptions*. The fall in Ecuadorian exposed employment already existed without a shock (-0.383 , $p < 0.10$, and more intense at earlier false dates): the negative Ecuadorian aggregate reads as secular drift, consistent with its leave-one-out fragility and with the SDID, which absorbs that drift by construction and eliminates it in placebo and real period alike (Table A11). Its contractionary informal channel, the evidence for Narrative C, does not appear in the placebo (-0.633 , n.s.). The Uruguayan exception, in contrast, survives: its exposed employment was not falling before the shock ($+0.022$, n.s., stable across the three false dates), so the real contraction (-0.163) belongs to the post-ChatGPT period. Brazil and Bolivia fit neither category cleanly and demand a reading of their own. The Brazilian false TWFE is contaminated in all three outcomes (employment $+0.163$, $p < 0.10$; wage $+0.061$, $p < 0.05$; hours $+1.177$, $p < 0.01$), which would recall the Ecuadorian case; but the pre-trends Wald does not reject employment parallelism ($p = 0.622$) and, above all, the false synthetic DiD is clean in all three ($+0.030$, $+0.010$, $+0.207$, all n.s.), so the naive

Table 10: Pre-COVID placebo: fake shock in 2018Q1 on the 2015–2019 vintages, same battery as the actual design

Country		log employment	log monthly wage	weekly hours
Bolivia	Fake $\hat{\tau}$ (TWFE)	−0.204	0.037	0.595
	Pre-trends Wald (p)	0.032**	0.771	0.776
	Fake $\hat{\tau}$ (SDID)	− 0.117 **	−0.018	0.558
Brazil	Fake $\hat{\tau}$ (TWFE)	0.163 *	0.061 **	1.177 ***
	Pre-trends Wald (p)	0.622	0.021**	0.052*
	Fake $\hat{\tau}$ (SDID)	0.030	0.010	0.207
Costa Rica	Fake $\hat{\tau}$ (TWFE)	−0.003	0.124	2.218
	Pre-trends Wald (p)	0.017**	0.016**	0.500
	Fake $\hat{\tau}$ (SDID)	0.103	0.024	−0.760
Ecuador	Fake $\hat{\tau}$ (TWFE)	− 0.383 *	0.037	2.414
	Pre-trends Wald (p)	0.002***	0.001***	0.065*
	Fake $\hat{\tau}$ (SDID)	0.055	−0.017	−0.106
Mexico	Fake $\hat{\tau}$ (TWFE)	0.054	− 0.072 ***	−0.170
	Pre-trends Wald (p)	0.196	0.000***	0.005***
	Fake $\hat{\tau}$ (SDID)	−0.007	− 0.050 ***	− 0.324 **
Peru	Fake $\hat{\tau}$ (TWFE)	0.025	−0.075	0.294
	Pre-trends Wald (p)	0.000***	0.001***	0.008***
	Fake $\hat{\tau}$ (SDID)	0.054	0.018	0.143
Uruguay	Fake $\hat{\tau}$ (TWFE)	0.022	0.018	1.342 ***
	Pre-trends Wald (p)	0.000***	0.000***	0.122
	Fake $\hat{\tau}$ (SDID)	0.015	0.020	0.421 **

Notes: Falsification on the same harmonized surveys over the 2015Q1–2019Q4 window (20 quarters, same length as the actual panel), with a fictitious launch of generative AI in November 2017 (fake post from 2018Q1; 12 pre and 8 post quarters). Each row replicates the specification of the actual design: the continuous TWFE of Table 5, the joint Wald test of the pre-period event-study coefficients, and the binary synthetic DiD of Table A11 (2017Q4 buffer excluded, mirroring the 2022Q4 buffer). SDID inference by normal approximation with the estimator’s placebo standard error. Colombia is excluded because the GEIH prior to the 2021 redesign codes occupation with the 2-digit CNO-70, insufficient to assign the β dose (the same reason its actual panel starts in 2022). Coefficients significant at 10 percent in bold. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TWFE contamination looks like a level artifact, not a directional trend, and one synthetic DiD absorbs by construction; the Brazilian employment reading therefore rests on synthetic DiD, whose aggregate binary estimate is +0.049 ($p < 0.05$; Table A11), more than on the aggregate TWFE. Bolivia exhibits the inverse pattern: its false TWFE is clean in all three outcomes (−0.204, +0.037, +0.595, all n.s.) and the pre-trends Wald only rejects in employment ($p = 0.032$), but the false synthetic DiD detects a significant employment effect (−0.117, $p < 0.05$) exactly where the real design is practically null (+0.011; Table A11), the exercise’s most concrete warning about the estimator’s precision in the sample’s smallest panel (59 treated occupations, 35 controls). No estimator is clean in both new countries at once, which reinforces rather than weakens reporting the three diagnostics in parallel. Third, the drifts the placebo does detect fall in cells where the paper claims nothing: the Mexican

wage shows a stable negative drift (-0.072 , $p < 0.01$, at any false date) that the SDID only partially absorbs, and Uruguayan hours a positive one ($+1.342$, $p < 0.01$); the corresponding real estimates are not significant and the text does not interpret them.

The placebo also provides a reading key for the body’s pre-trends tests. In several country×outcome cells the placebo Wald rejects parallelism (the cell panels are noisy in 2015–2019 as well) and yet the accumulated false effect is zero: the violations it detects are mean-zero oscillations, not directional drifts that accumulate bias. That distinction, invisible to the joint test, is exactly what separates a contaminated design from a noisy one, and Table 11 formalizes it as an empirical calibration of the bounds of Rambachan and Roth (2023). Three numbers summarize it. First, the maximum historical quarterly violation of parallelism (v^{hist} , measured over the full placebo, where any drift counts as a violation since there is no shock) is of the same order as the real pre-period’s: the ratio $\varphi = v^{\text{hist}}/v^{\text{pre}}$ ranges from 0.5 to 3.1 with median 1.1, so the real period’s imperfect pre-trends are the ordinary weather of these panels, not a sign of anticipation. Second, and in honesty: at $\bar{M} = \varphi$ exactly, the relative-magnitude bounds are inconclusive in all twenty-one cells, with robust intervals of ± 0.4 to ± 37 ; with twelve post quarters, the violations model allows deviations of historical size to accumulate in the same direction for three years, a trajectory these panels’ history never produced. Third, the final column measures the directional accumulation that history did produce: the false $\hat{\tau}$, the very statistic those bounds constrain, stays far below what they tolerate in all cells (in the worst case, the Ecuadorian drift of -0.383 against an interval of ± 1.5) and is approximately zero where pattern P lives (Costa Rica -0.003 and Peru $+0.025$, against intervals of ± 2.8 and ± 1.7). The joint reading: with a short pre-period and a long post-period, relative-magnitude bounds are nearly vacuous by construction, and the operative falsification of the identifying assumption is the placebo, which tests the realized accumulation, not the permitted one, over a window of identical length; the preregistered criterion $\bar{M} \geq 2$ remains reported for transparency.

Figure 3 shows the seven placebo event studies for employment: the absence of a break at 2018Q1 in Costa Rica, Peru, Mexico, and Uruguay, and the Ecuadorian secular slope, are visible to the eye; the Brazilian and Bolivian panels match the text’s mixed reading: no sharp break at 2018Q1 in either, but the sample’s largest dispersion in the Bolivian panel, a reflection of its reduced size.

The falsification also has to reach the estimates on which the article rests, and until now it did not: the exercise above is built on aggregate cells and on the interaction channels, not on the sector cells of Section 5.1. Table 12 closes that gap by rebuilding those cells on the 2015–2019 vintages, with the same harmonization and the same formality rule, and re-estimating the same equation against the fictitious launch. The central result passes in five of the six countries where the formal wage falls. In Bolivia, Brazil, Costa Rica, Ecuador and Peru the fake coefficient is indistinguishable from zero, and in Costa Rica and Ecuador it even carries the opposite sign, which makes the real estimate conservative. Mexico is the exception and an unambiguous one: its fake formal wage effect is -0.085 ($p < 0.001$), larger in magnitude than the real -0.059 , and it repeats almost identically across the three fictitious dates (-0.085 , -0.081 , -0.087). What the sector-specific estimate recovers in Mexico is a secular drift in the wages of exposed formal occupations that predates generative AI, so the Mexican cell is reported but not counted, and the article claims five of the eight rather than

Placebo event study of (log) employment: fake 2018Q1 shock, ref. 2017Q4

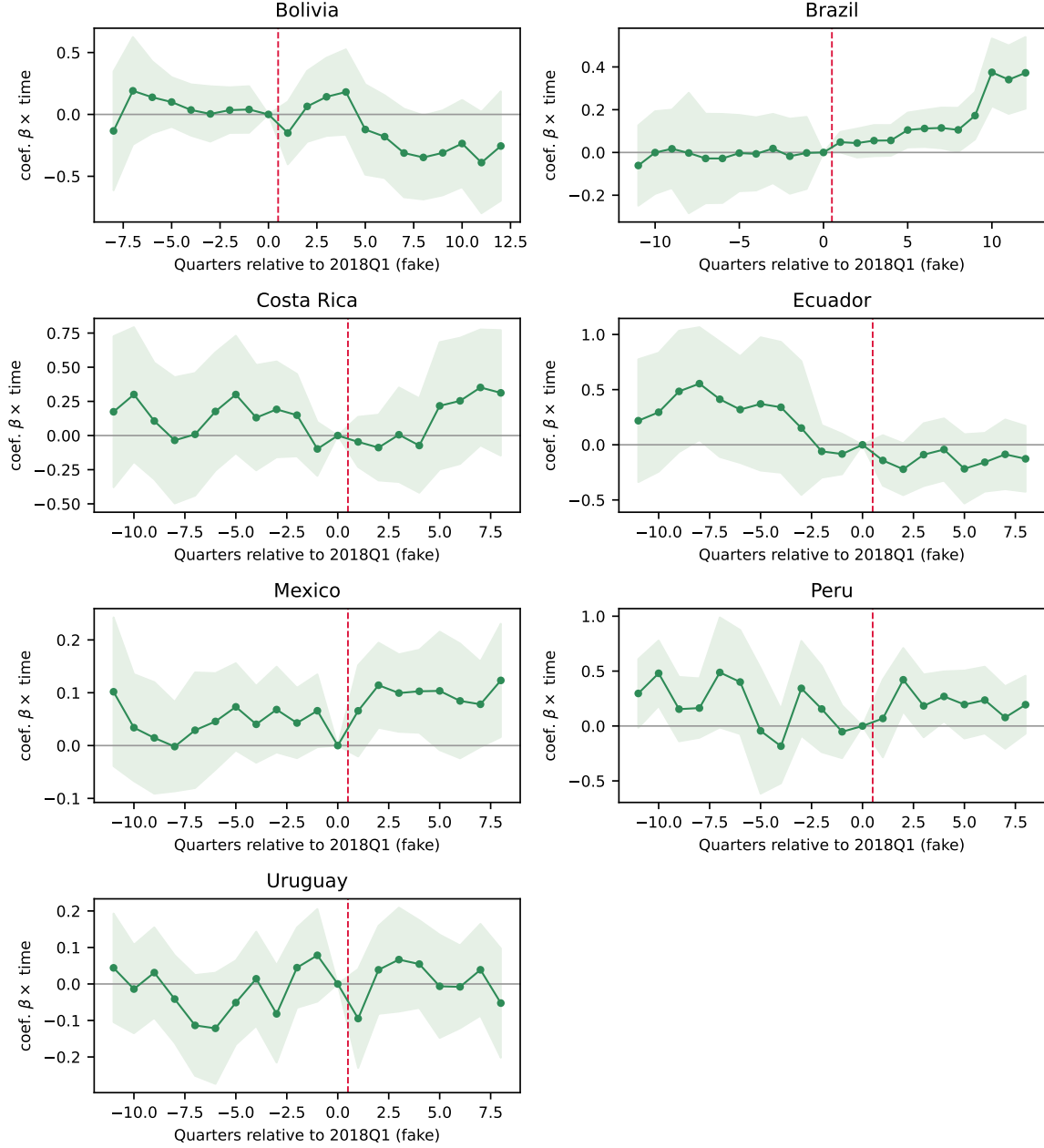


Figure 3: *Placebo event study of employment (log) by country.* Coefficients on the $\beta_o \times \text{time}$ interaction from the continuous-treatment event study on the 2015Q1–2019Q4 vintages, with a false shock in 2018Q1 (dashed line) and reference 2017Q4. Same conventions as Fig. 2. Colombia is excluded for the granularity of its occupational classifier prior to the GEIH redesign.

Table 11: Empirical calibration of $\bar{M}$ with the pre-COVID placebo: historical violations of parallelism versus their realized accumulation

	Actual v^{pre}	Placebo v^{hist}	$\varphi = v^{\text{hist}}/v^{\text{pre}}$	Robust CI at $\bar{M} = \varphi (\pm)$	Fake $\hat{\tau}$
<i>log employment</i>					
Bolivia	0.114	0.324	2.84	1.92	-0.204
Brazil	0.093	0.202	2.17	0.60	0.163*
Costa Rica	0.359	0.290	0.81	2.78	-0.003
Ecuador	0.144	0.211	1.47	1.51	-0.383*
Mexico	0.088	0.068	0.78	0.76	0.054
Peru	0.289	0.526	1.82	1.67	0.025
Uruguay	0.193	0.134	0.69	1.62	0.022
<i>log monthly wage</i>					
Bolivia	0.128	0.116	0.91	0.75	0.037
Brazil	0.050	0.048	0.95	0.40	0.061**
Costa Rica	0.205	0.172	0.84	1.37	0.124
Ecuador	0.220	0.266	1.21	1.18	0.037
Mexico	0.057	0.062	1.08	0.58	-0.072***
Peru	0.435	0.247	0.57	1.03	-0.075
Uruguay	0.154	0.174	1.13	1.15	0.018
<i>weekly hours</i>					
Bolivia	1.275	3.998	3.13	18.80	0.595
Brazil	0.635	0.937	1.48	4.73	1.177***
Costa Rica	3.642	4.367	1.20	21.03	2.218
Ecuador	1.402	3.957	2.82	22.33	2.414
Mexico	1.333	1.341	1.01	6.36	-0.170
Peru	3.027	4.557	1.51	37.27	0.294
Uruguay	2.400	1.162	0.48	12.85	1.342***

Notes: v^{pre} : maximum violation of parallelism in the actual pre-period (maximum absolute value of the difference between consecutive event-study coefficients, including the arrival at the reference), the denominator of the relative-magnitude bounds of Rambachan and Roth (2023). v^{hist} : the same magnitude over the full 2015–2019 placebo event study. φ : the $\bar{M}$ it would suffice to tolerate if post-2023 violations resembled the historical ones. Fifth column: half-width of the robust CI for the average post effect at $\bar{M} = \varphi$; last column: the fake $\hat{\tau}$ from the placebo TWFE (Table 10), the realized historical accumulation. Colombia is excluded from the placebo. Fake coefficients significant at 10 percent in bold. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

six. Two further readings follow from the same table. The Brazilian informal channel is contaminated in all three outcomes (fake employment +0.454, wage +0.134, hours +1.995, all significant), so the Brazilian informal wage spillover, which the interaction design reports as its most marked, must be read with the same caution as the Mexican formal wage. And hours fail the placebo in several countries, consistent with their being the least reliable margin throughout the article. Applying the Benjamini-Hochberg correction within the family of the 48 sector contrasts, the surviving cells at the 10 percent level are those marked in the last column: the formal wage in Costa Rica, Ecuador and Peru among the countries that also clear the placebo.

Table 12: Falsifying the central finding: pre-COVID placebo on the sector cells

Country	Sector	Employment (log)		Monthly wage (log)		BH
		Real	Fake	Real	Fake	
Bolivia	Formal	−0.056	−0.375*	−0.070**	−0.005	−
	Informal	+0.065	−0.056	−0.004	+0.065	−
Brazil	Formal	+0.019	+0.119	−0.047**	+0.002	−
	Informal	+0.213***	+0.454***	+0.067**	+0.134***	E
Costa Rica	Formal	+0.207*	+0.040	−0.109**	+0.028	W
	Informal	+0.266**	+0.157	−0.063	+0.345**	−
Ecuador	Formal	+0.147	−0.167	−0.232***	+0.080	W
	Informal	−0.286**	−0.371	+0.091*	+0.118	−
Mexico	Formal	+0.007	−0.004	−0.059***	−0.085***	W
	Informal	+0.068	+0.064	−0.018	−0.029	−
Peru	Formal	+0.152**	−0.020	−0.082**	−0.043	W
	Informal	+0.177	+0.162*	−0.060	−0.036	−
Uruguay	Formal	−0.213***	+0.017	−0.001	−0.026	E
	Informal	+0.102	−0.004	−0.118	−0.092**	−

Notes: “Real” reproduces Table 4. “Fake” estimates the same equation on sector cells rebuilt from the 2015–2019 vintages with a fictitious launch in 2018Q1, using the same harmonization, the same formality rule and the same minimum cell size. Colombia is excluded for lack of comparable pre-COVID vintages, as in Table 10. The BH column reports which cells survive the Benjamini-Hochberg correction at 10 percent within the family of 48 sector contrasts (E: employment; W: wage). A significant fake coefficient of the same sign as the real one indicates secular drift and contaminates the reading; one of the opposite sign makes the real estimate conservative. Hours are omitted from the table and discussed in the text. Occupation-clustered standard errors. Coefficients significant at 5 percent in bold. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6 Conclusions

This paper delivers country-by-country causal estimates $\{\hat{\tau}^c\}_{c=1}^8$ for Costa Rica, Colombia, Ecuador, Mexico, Peru, Uruguay, Brazil, and Bolivia, built on panels harmonized to four-digit ISCO-08 with the complementary exposure β of Eloundou et al. (2024) as treatment intensity. The side-by-side comparison, rather than a pooled coefficient, is the central empirical contribution.

Three conclusions emerge from the comparison. First, aggregate effects are heterogeneous and, in several countries, small or statistically indistinguishable from zero: aggregate employment is positive and significant in Costa Rica, Peru-ENAH, and Brazil, negative and significant in Ecuador and Uruguay, and not distinguishable from zero in Colombia, Mexico, and Bolivia. Read in isolation, several of these aggregates would suggest a replication of the

null of Humlum and Vestergaard (2025a).

Second, those aggregates conceal the systematic pattern, which appears once outcomes are separated by formality. The pattern is a price rather than a quantity: what generalizes across the sample is the decline of the formal wage, not the movement of formal employment. LLM exposure tends to reduce monthly wages, and more weakly to expand employment, in occupations with a predominantly formal baseline, sharpest in Costa Rica, Peru, and Brazil (positive formal employment, negative formal wages, significant in all three). Rebuilding the cells separately for formal and informal workers, so that the outcome is each sector’s own rather than the occupation total, sharpens this reading and splits it in two (Section 5.1): the fall in the formal wage is confirmed by direct estimation in six of the eight countries, five of which clear the placebo built on the same cells, and three of those five (Costa Rica, Ecuador, and Peru) also clear the multiple-testing correction within the sector-specific family; Mexico is the exception, since there the fictitious launch reproduces a larger decline than the real one, whereas the expansion of formal employment survives only in Peru and, marginally, Costa Rica. The informal channel is weaker and less systematic in sign, except in Brazil, where the informal wage is positive and significant; the composition test attributes that spillover to selective retention of informal incumbents, not a generalized wage improvement, and the sector-specific placebo further flags it as contaminated, so it is reported with the same caution as the Mexican wage cell. The Colombian and Mexican nulls differ in nature: in Colombia employment combines opposite-signed but imprecise channels, with demonstrable cancellation only in hours, whereas in Mexico both channels are indistinguishable from zero and the null reads as a genuine absence of effect (Narrative B); Bolivia approximates the Colombian case, with formal employment of pattern P’s order of magnitude but without conventional significance, and a formal wage that does fall significantly (-0.092 , $p < 0.05$). Baseline formality as the central dimension of heterogeneity is thus empirically justified.

Third, neither channel is universal. In the wage channel, where the central finding rests, Colombia is the only sign exception and Uruguay sits at a precise zero. In employment, Uruguay reverses the formal channel’s sign (from positive to negative and significant), and Ecuador departs by a different route: its formal employment channel is null and the negative effect concentrates in the informal channel. These deviations suggest that institutional mediation, and not only the technical capability of LLMs, governs the labor market response, in line with automation evidence in which the generosity of unemployment insurance cushions the wage effects of robots (Brollo, 2024); Section 5.4 discusses them alongside the interpretive Narratives.

Where that price adjustment happens is what gives the article its mechanism, and it also reconciles the finding with the OECD evidence. The decline in the formal wage is located among the recently hired rather than spread over established workers: the entrant-incumbent differential is negative and significant in all five countries with tenure data. How much of the decline the established bear varies: in Peru and Brazil incumbent earnings do not move at all, in Costa Rica they fall by about two fifths of what entrants lose (-0.082 against -0.204), and in Ecuador by half (-0.210 against -0.417), so in those two countries the adjustment is only partly compositional. Read that way, the precise null that Humlum and Vestergaard (2025a) report for Danish workers and the wage decline reported here are not in conflict, since in the countries where the adjustment is entirely compositional incumbent earnings are unmoved

here too; what a segmented labor market with a flexible entry wage adds is a margin on which the shock can land, and which a market that adjusts through hiring quantities would express elsewhere. Subordinate to that price finding, and exploratory because it was articulated after observing the Peruvian data, stands a fifth pattern none of the four Narratives anticipates: in the interaction specification, exposed formal-dominant occupations combine downward pressure on monthly wages with an employment expansion that direct estimation narrows to Peru and, marginally, Costa Rica, while informal-dominant ones show no employment movement statistically distinguishable from zero, plus rising wages among those who remain. Brazil, with a different sample and survey, reproduces that interaction signature as sharply as Costa Rica and Peru, and Bolivia in a weaker version (significant formal wage, formal employment of the right sign but imprecise), so the wage half of the pattern no longer rests on a single country or a single survey. The individual-level composition test of Section B identifies the mechanism behind the first half: the formal wage decline reflects lower entry wages, with incumbent wages intact (in Peru, an exact zero; Brazil replicates the mechanism with an even larger entry compression), that is, recomposition rather than income loss among the established in those two countries. This fits a skill-biased complementarity channel in the formal sector, modulated by structural segmentation, plus a recomposition channel in the informal sector whose mechanism repeated cross-sections cannot identify. The contribution thus shifts from “which narrative each country supports” to “formal-informal segmentation shifts the LLM shock onto an entry price margin the first OECD-centered causal estimates were not positioned to observe”.

The entry margin invites a sharper comparison with the US evidence. Brynjolfsson, Chandar, and Chen (2025) document, in administrative payroll data, that employment losses in exposed occupations concentrate among early-career workers: the US entry margin adjusts through quantities (fewer young hires), with limited movement in compensation. My composition test finds the same margin adjusting through prices in Latin America: exposed formal occupations absorb entrants at lower entry wages. The quantity margin moves here too, and in the same direction: the entrant share falls significantly in four of the five countries, so the contrast is not price against quantity but the addition of a price margin that the US evidence does not find. An institutional prediction follows that future work can test directly: where entry wages are flexible and unemployment insurance is weak, the LLM shock should surface in entry prices; where hiring is the freer margin and wage floors bind, in hiring quantities.

Limitations and work in progress. Four limitations guide the next version. First, the Honest-DiD bounds, run on the aggregate estimates (Table 8) and on the channel coefficients $\hat{\tau}_F$ and $\hat{\tau}_I$ separately (Table 9), meet the self-imposed criterion $\overline{M} \geq 2$ in no country, so inference robust to deviations from parallel trends is fragile for the aggregate and for the channels on which the central finding rests (the empirical calibration of Section 5.7 sharpens the diagnosis: with a short pre-period and twelve post quarters, those bounds allow directional accumulations without precedent in the panels’ own history, so they are nearly vacuous by construction and the operative falsification is the placebo); the pre-trends of the formal employment channel, moreover, reject parallelism in Peru, Uruguay, Brazil, and Bolivia, so the reading of the fifth pattern in light of the four Narratives remains exploratory. The heterogeneity specification carries a further caveat: it omits the lower-order interaction

between baseline formality and the post indicator, and once it is included both the formal employment coefficient in Costa Rica, Peru and Brazil and the formal wage coefficient in Costa Rica, Peru and Bolivia are absorbed, the latter surviving within that specification only in Brazil and Ecuador (Table A13), so within that design an exposure effect concentrated in formal occupations cannot be separated from a post-2022 shift specific to formality. The sector-specific estimates are reported precisely because they do not depend on it. Second, outcomes are computed conditional on being employed, so selection through occupational migration and exit from employment is not directly identifiable with repeated cross-sections. Third, the exposure measures of Webb (2019) (tasks and patents) and Felten et al. (2021) (abilities) remain pending: they require external occupational mappings unavailable here (the alternative measures α and ζ of Eloundou et al. (2024) already appear as robustness in Table A3); so does the sensitivity of the classifier harmonization to alternative weighting schemes. Fourth, the two sign exceptions (Ecuador and Uruguay) stand differently after the pre-COVID placebo of Section 5.7: the Ecuadorian one continues a pre-shock trend and is reinterpreted as secular, while the Uruguayan one does not appear in the no-shock world and requires finer institutional examination linking the reversal of the formal channel to its labor regime. The same exercise leaves two cautions on Brazil and Bolivia: the naive TWFE placebo is contaminated in Brazil (though synthetic DiD clears it), and synthetic DiD detects a significant false employment effect in Bolivia, the sample’s smallest panel, so the Bolivian employment result is reported but not interpreted as robust under any estimator.

Read jointly, the eight countries show that the labor market effect of LLMs in Latin America surfaces as a price before it surfaces as a quantity: the formal monthly wage of exposed occupations falls in six of the eight countries when estimated directly on formal workers, and in five of them the decline survives a placebo on the same cells, while employment moves little and in country-specific ways, and that adjustment is located in the entry wage, with established workers’ earnings intact in Peru and Brazil and only partly affected in Costa Rica and Ecuador. Read that way, the result does not contradict the null of the first OECD estimates; it shows which margin the shock lands on when the labor market is segmented and the entry wage is flexible. That margin, and not the aggregate effect, is the paper’s central finding.

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Appendices

A Additional tables and figures

This appendix gathers the tables and figures referenced from the body: [the channels diagram \(Fig. A1\)](#), the digitalization and adoption context by country ([Table A1](#)), the event studies for the wage and hours ([Figs. A2 and A3](#)), the binary Callaway–Sant’Anna estimator ([Table A2](#)) and the alternative exposure measures with the IV contrast for dose measurement error ([Table A3](#)).

The event studies for the other two outcomes confirm the Wald test’s credibility hierarchy: the monthly wage ([Fig. A2](#)) is noisy pre-treatment and deviates from zero before 2023Q1 in several countries, which is why the aggregate wage reading is the least reliable; weekly hours ([Fig. A3](#)) sit in between.

The IV rows of [Table A3](#) attack measurement error in the dose: β and ζ aggregate the same latent exposure with different rubric weightings, so instrumenting $\beta_o \times \text{post}_t$ with $\zeta_o \times \text{post}_t$ purges the weighting-specific error (not the crosswalk chain’s common error; the contrast is a lower bound on the attenuation correction). The instrument is very strong

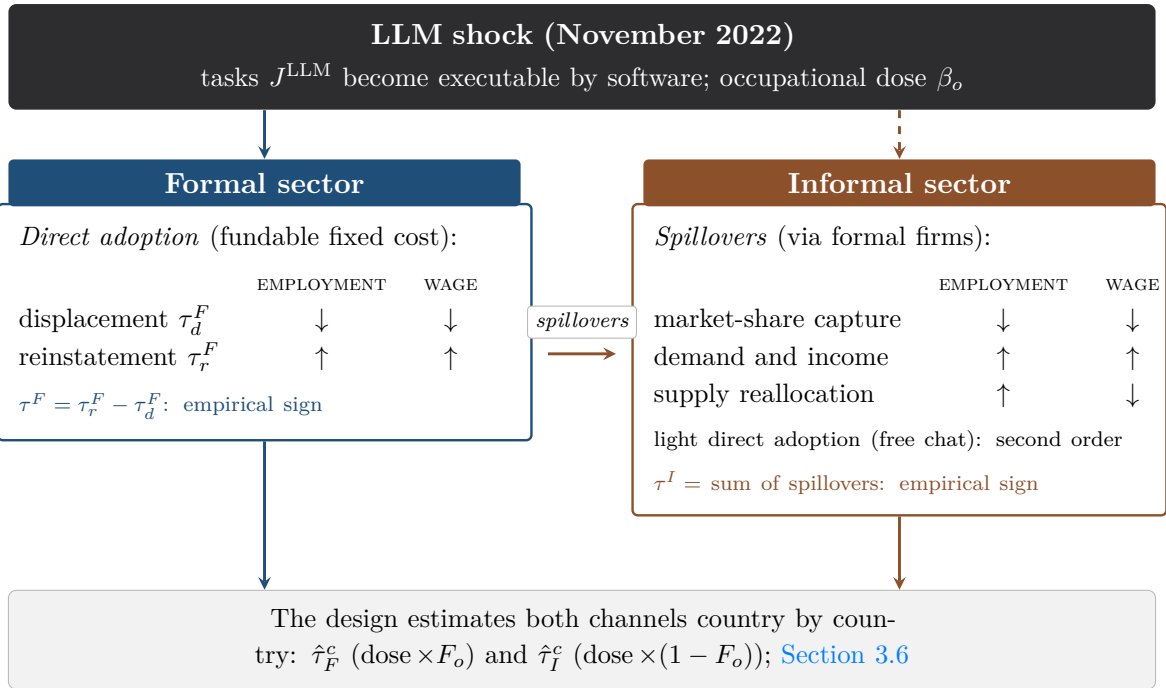


Figure A1: *Channels of the LLM shock on the formal and informal sectors.* The formal sector adopts directly (fundable fixed cost) and faces the displacement and reinstatement channels; the informal sector receives the shock mainly via three general-equilibrium spillovers whose signs on employment and the wage differ, plus a light direct adoption of second order (dashed arrow). The $\uparrow/\downarrow$ arrows give each channel’s direction on the sector’s employment and wage. See [Section 2.3](#).

Table A1: Digitalization, generative AI adoption, and informality by country

Country	Internet use (%)	Fixed broadband	Mobile	Claude per capita	ChatGPT rank	Baseline informality (%)
Bolivia	72.6	11.0	101.6	0.53	59	68.1
Brazil	80.5	21.0	101.2	0.91	45	38.8
Colombia	72.8	17.0	156.2	1.14	49	62.2
Costa Rica	82.6	21.8	136.5	1.56	36	44.5
Ecuador	69.7	15.1	98.1	0.47	45	73.2
Mexico	78.6	20.6	105.7	0.61	59	56.5
Peru	74.7	9.5	124.1	0.96	44	71.6
Uruguay	89.9	33.5	139.8	1.83	—	23.4

Notes: Internet use: individuals using the internet as a percentage of the population. Fixed broadband and mobile subscriptions: per 100 inhabitants. All three indicators come from the World Bank’s World Development Indicators (ITU origin), year 2022 (accessed: July 2026). Claude use per capita: index from the Anthropic Economic Index (Anthropic, 2026), April–May 2026 average, where 1 equals usage proportional to the country’s working-age (15–64) population. ChatGPT rank: the country’s position in the ranking of ChatGPT messages per capita among 119 countries, 2025Q1–2026Q1 average, public data accompanying Chatterji et al. (2025); Uruguay does not appear in that panel. Baseline informality is that of Table 3. Exposure β is common across countries by construction; these indicators describe the realized-adoption margin that scales the reduced form $\hat{\tau}^c$ and organize the reading of prediction P3.

(occupational correlation 0.96 to 0.97; first-stage F between 492 and 1,708) and the result clear-cut: all eleven significant aggregate estimates retain their sign, ten of eleven at the 5 percent level, and the IV/OLS ratios run from 0.68 to 1.28 with no systematic amplification, ruling out first-order attenuation.

B Additional robustness

The post-COVID rebound threatens the sample window: with the baseline in 2021, a differential recovery of high-exposure cognitive occupations (more teleworkable, harder hit sectorally) could be confounded with the launch. Table A4 evaluates two corrections on the aggregate. Dropping 2021 from the pre-period leaves signs and significance essentially intact, with three exceptions that weaken: Ecuadorian employment, the Peruvian wage, the Bolivian wage. Sector-by-quarter fixed effects are more demanding: employment and wage aggregates attenuate across the board, so much of the aggregate variation co-moves with sectoral trajectories. The decomposition by formality, however, survives within sector (Table A5): the Peruvian formal channel retains its full signature (employment +0.269, wage -0.123 , both $p < 0.05$), the formal wage keeps falling in Costa Rica, Ecuador, and Bolivia, the Uruguayan formal contraction remains (-0.207 , $p < 0.01$), and Bolivian formal employment gains significance (+0.188 n.s. to +0.328, $p < 0.05$, the only channel that strengthens under the control). Costa Rican formal employment and Brazilian formal employment and wage attenuate to insignificance, though the Brazilian informal wage, the country’s distinctive signature, remains intact (+0.199, $p < 0.05$). The differential sectoral recovery thus

Event study of (log) monthly wage by country: $\beta_o \times$ time interaction, ref. 2022Q4

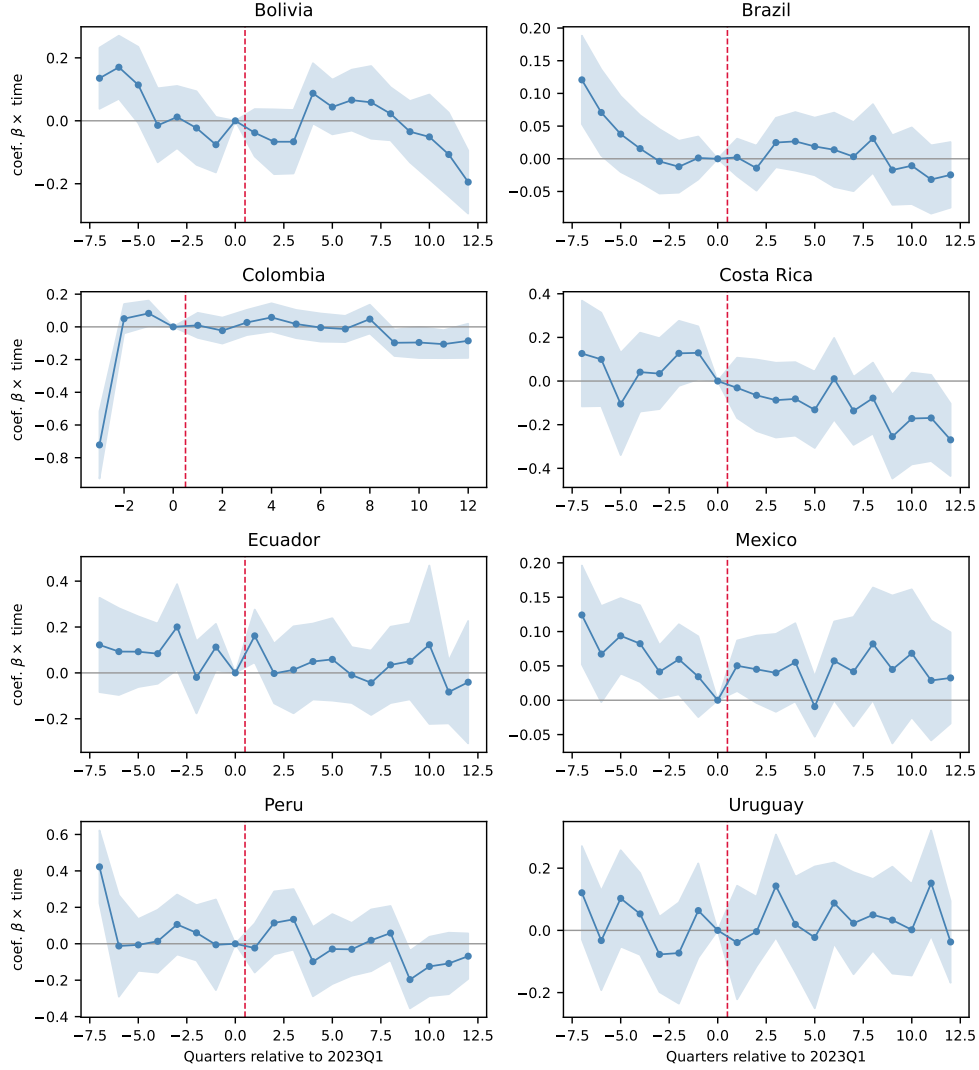


Figure A2: *Event study of the monthly wage (log) by country.* Coefficients on the $\beta_o \times$ time interaction from the continuous-treatment event study, same specification and conventions as Fig. 2. The non-zero pre-treatment trajectories are consistent with the rejected Wald test of wage parallelism in seven of the eight countries.

contaminates the aggregates above all; the formality decomposition's signal operates within sectors.

A second check examines the definition of formality, whose second criterion, tax registration for the self-employed, is the noisier across surveys; Table A6 redefines the baseline formality share F_o using only wage earners. The picture survives the redefinition, and the formal wage channel, on which the central finding rests, holds up best: formal employment still expands in Costa Rica (+0.357, $p < 0.01$), Peru (+0.442, $p < 0.01$), Mexico (+0.104, $p < 0.10$) and Brazil (+0.170, $p < 0.01$), the formal wage still falls in Costa Rica (−0.223,

Event study of weekly hours by country: $\beta_o \times$ time interaction, ref. 2022Q4

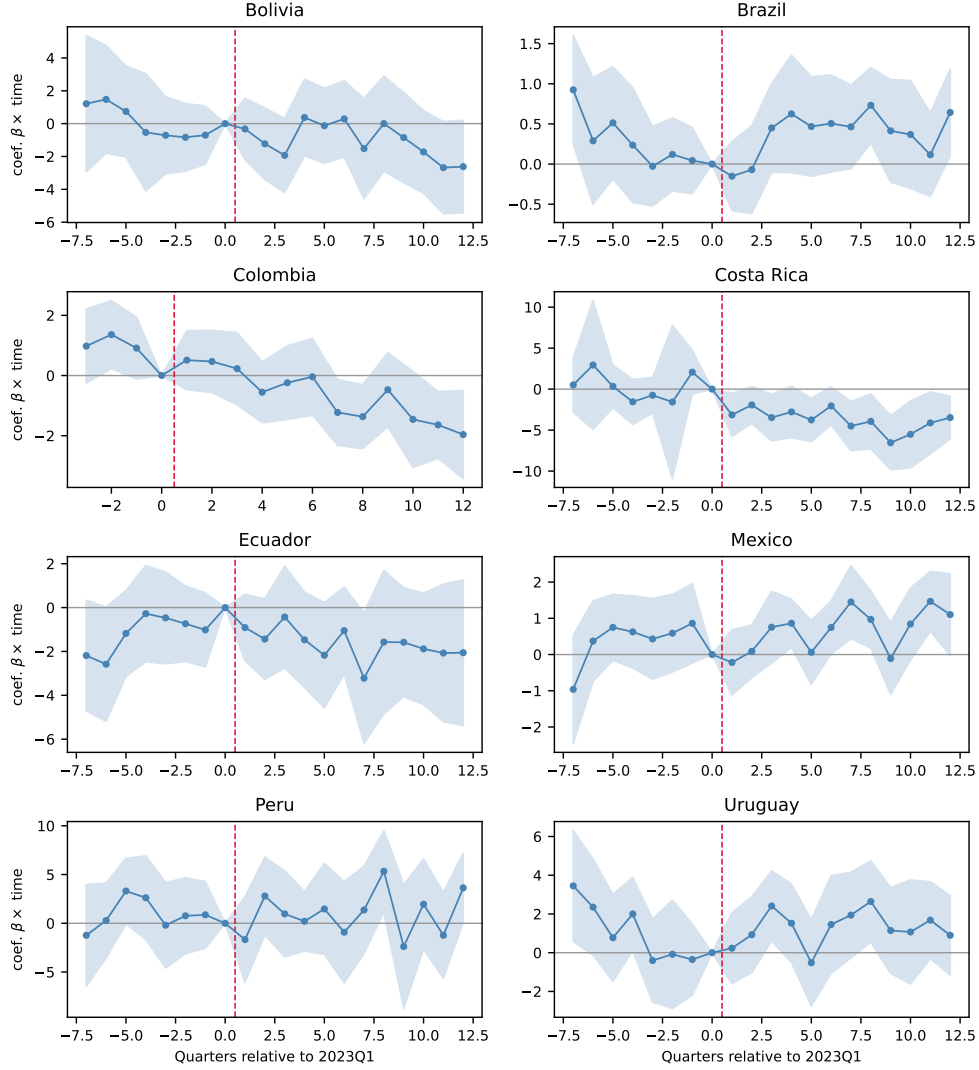


Figure A3: *Event study of weekly hours by country.* Coefficients on the $\beta_o \times$ time interaction from the continuous-treatment event study, same specification and conventions as Fig. 2.

$p < 0.01$), Peru (-0.136 , $p < 0.01$), Ecuador (-0.130 , $p < 0.01$), Brazil (-0.061 , $p < 0.01$), and Bolivia (-0.077 , $p < 0.05$), and Uruguayan formal employment still contracts (-0.165 , $p < 0.05$); Bolivian formal employment still misses conventional significance ($+0.141$, n.s.). Changes concentrate in the informal channel, more sensitive to the definition: the Colombian informal wage loses significance ($+0.101$ to $+0.040$) and Uruguayan informal employment, already imprecise, remains so ($+0.216$ to $+0.172$), while the Ecuadorian informal wage intensifies ($+0.364$, $p < 0.01$), as do the Brazilian informal wage ($+0.202$ to $+0.222$, both $p < 0.01$) and the Bolivian contraction of informal hours (-3.393 to -4.109 , from $p < 0.10$ to $p < 0.05$).

A third question is whether one major occupational group drives the results, since the

Table A2: Binary Callaway–Sant’Anna estimator (high vs. low baseline exposure) by country

		log employment	log monthly wage	weekly hours
Bolivia	$\widehat{ATT}$	−0.023	0.002	−0.431
	95% CI	[−0.109, 0.062]	[−0.027, 0.031]	[−1.095, 0.233]
Brazil	$\widehat{ATT}$	0.056	0.006	0.095
	95% CI	[0.010, 0.101]	[−0.014, 0.026]	[−0.128, 0.317]
Colombia	$\widehat{ATT}$	0.010	−0.026	−0.333
	95% CI	[−0.086, 0.106]	[−0.062, 0.011]	[−0.743, 0.078]
Costa Rica	$\widehat{ATT}$	0.080	0.040	−1.703
	95% CI	[−0.091, 0.252]	[−0.055, 0.134]	[−3.272, −0.134]
Ecuador	$\widehat{ATT}$	−0.044	0.031	−0.483
	95% CI	[−0.132, 0.044]	[−0.028, 0.090]	[−1.286, 0.320]
Mexico	$\widehat{ATT}$	0.059	0.011	0.319
	95% CI	[−0.004, 0.122]	[−0.010, 0.032]	[−0.057, 0.695]
Peru	$\widehat{ATT}$	0.068	0.047	0.582
	95% CI	[−0.025, 0.161]	[−0.016, 0.111]	[−0.941, 2.105]
Uruguay	$\widehat{ATT}$	−0.106	−0.019	0.344
	95% CI	[−0.242, 0.031]	[−0.081, 0.042]	[−0.557, 1.246]

Notes: Global ATT of Callaway and Sant’Anna (2021) with occupations whose β exposure is above the base-year employment-weighted median as the treated group and low-exposure occupations as the never-treated control, treatment in 2023Q1. Bootstrap 95% confidence intervals. Effects whose CI excludes zero in bold.

estimator weights by employment. Table A7 re-estimates the aggregate $\hat{\tau}^c$ omitting each major group of the native classifier (first digit) in turn, and reports the range. In six of the eight countries the employment coefficient never changes sign, and the significant estimates for Costa Rica, Peru, Brazil and Uruguay hold in almost all omissions (at most one exception out of nine; Brazil is significant at the 5 percent level in all nine). The least stable case is Ecuador: its negative employment coefficient keeps its sign in all nine omissions but loses 5 percent significance in five, so the Ecuadorian aggregate demands caution beyond Honest-DiD’s; its negative informal channel, in contrast, survives the other checks here. The two exceptions to the sign rule are Bolivia and Colombia, whose employment coefficients, never significant in the baseline, cross zero in some omissions (Bolivia between −0.046 and +0.161; Colombia between −0.059 and +0.042), consistent with already imprecise results; the Bolivian monthly wage, in contrast, remains negative in all nine omissions and significant at the 10 percent level or better in five, the most stable piece of the Bolivian row.

A fourth question is whether the formality margin actually captures a skill margin, since informal workers tend to be less skilled. At the occupation level a *horse race* answers in part: Table A8 interacts the dose simultaneously with the baseline shares of formality (F_o)

Table A3: Robustness to the exposure measure: aggregate estimates $\hat{\tau}^c$ under alternative measures from Eloundou et al. (2024) and IV for measurement error

		log employment	log monthly wage	weekly hours
Bolivia	α (direct)	0.324*	-0.031	-0.025
	β (complementary)	0.069	- 0.059*	-1.067
	ζ (upper bound)	0.075	- 0.040*	-0.652
	β , IV with ζ	0.103	- 0.055*	-0.893
Brazil	α (direct)	0.163***	- 0.068*	0.298
	β (complementary)	0.123***	-0.025	0.135
	ζ (upper bound)	0.084***	-0.024	0.139
	β , IV with ζ	0.117***	-0.034	0.194
Colombia	α (direct)	0.217	-0.022	- 1.983***
	β (complementary)	-0.017	0.118***	- 1.468***
	ζ (upper bound)	0.041	0.060**	- 1.209***
	β , IV with ζ	0.055	0.080**	- 1.612***
Costa Rica	α (direct)	0.356**	-0.070	0.494
	β (complementary)	0.302***	- 0.178***	- 4.027***
	ζ (upper bound)	0.228***	- 0.109***	- 2.212***
	β , IV with ζ	0.313***	- 0.149***	- 3.033***
Ecuador	α (direct)	-0.104	0.051	0.819
	β (complementary)	- 0.164**	-0.058	-0.611
	ζ (upper bound)	- 0.114*	-0.019	-0.305
	β , IV with ζ	- 0.155*	-0.026	-0.415
Mexico	α (direct)	0.041	0.047	1.038
	β (complementary)	0.064	-0.017	0.325
	ζ (upper bound)	0.056	-0.001	0.350
	β , IV with ζ	0.075	-0.002	0.468
Peru	α (direct)	0.403***	- 0.171***	0.667
	β (complementary)	0.283**	- 0.097**	0.135
	ζ (upper bound)	0.262***	- 0.095***	0.269
	β , IV with ζ	0.339***	- 0.123***	0.348
Uruguay	α (direct)	- 0.206*	0.025	0.086
	β (complementary)	- 0.163**	0.015	0.319
	ζ (upper bound)	- 0.131**	0.019	0.263
	β , IV with ζ	- 0.179**	0.026	0.359

Notes: Each cell is the coefficient $\hat{\tau}^c$ on the interaction ($\text{measure}_o \times \text{post}_t$) in the same TWFE specification of Table 5 (occupation and quarter fixed effects, weighting by expanded employment, errors clustered at the occupation level), varying only the exposure measure in the dose. Following the task-level rubric of Eloundou et al. (2024), α counts only directly exposed tasks; β adds half of the tasks exposed through software built on LLMs; and ζ counts them with full weighting, so it is the upper bound (and not the sum $\alpha + \beta$). The β row reproduces Table 5. The IV row instruments $\beta_o \times \text{post}_t$ with $\zeta_o \times \text{post}_t$ in the same specification, purging the measurement error specific to the rubric weighting (not the error common to the crosswalk chain, so it is a lower bound of the attenuation correction); first-stage F between 492 and 1,708. The measures of Webb (2019) and Felten et al. (2021) require external mappings not available in this version and remain pending. Effects significant at 10 percent in bold. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4: Robustness to the post-COVID rebound: excluding 2021 and with sector×quarter fixed effects

	Baseline specification	Excluding 2021	Sector×quarter
<i>log employment</i>			
Bolivia	0.069 (0.121)	0.073 (0.103)	0.225 (0.141)
Brazil	0.123*** (0.042)	0.123*** (0.034)	0.014 (0.047)
Colombia	−0.017 (0.089)	−0.017 (0.089)	−0.022 (0.102)
Costa Rica	0.302*** (0.098)	0.255*** (0.095)	0.069 (0.130)
Ecuador	− 0.164** (0.082)	−0.115 (0.078)	−0.072 (0.089)
Mexico	0.064 (0.049)	0.061 (0.044)	−0.007 (0.035)
Peru	0.283** (0.112)	0.241*** (0.078)	0.116 (0.152)
Uruguay	− 0.163** (0.073)	− 0.154** (0.074)	− 0.176* (0.096)
<i>log monthly wage</i>			
Bolivia	− 0.059* (0.030)	0.000 (0.031)	−0.029 (0.046)
Brazil	−0.025 (0.025)	0.005 (0.023)	0.018 (0.024)
Colombia	0.118*** (0.033)	0.118*** (0.033)	0.089** (0.037)
Costa Rica	− 0.178*** (0.050)	− 0.196*** (0.050)	−0.073 (0.060)
Ecuador	−0.058 (0.041)	−0.044 (0.044)	−0.069 (0.052)
Mexico	−0.017 (0.024)	0.012 (0.025)	−0.016 (0.015)
Peru	− 0.097** (0.049)	−0.070 (0.050)	−0.060 (0.061)
Uruguay	0.015 (0.027)	0.055* (0.031)	−0.027 (0.040)
<i>weekly hours</i>			
Bolivia	−1.067 (1.153)	−0.435 (0.801)	0.405 (1.272)
Brazil	0.135 (0.173)	0.353* (0.209)	−0.387 (0.279)
Colombia	− 1.468*** (0.356)	− 1.468*** (0.356)	− 1.140*** (0.433)
Costa Rica	− 4.027*** (1.130)	− 3.759*** (1.156)	− 2.362* (1.371)
Ecuador	−0.611 (0.741)	−1.102 (0.833)	− 1.737* (0.991)
Mexico	0.325 (0.478)	0.208 (0.476)	0.353 (0.334)
Peru	0.135 (0.675)	0.623 (0.858)	− 1.710* (0.923)
Uruguay	0.319 (0.451)	1.483*** (0.413)	0.025 (0.527)

Notes: The baseline column reproduces Table 5. “Excluding 2021” restricts the pre-period to 2022Q1–2022Q3, removing the quarters of sharpest post-COVID recovery. “Sector×quarter” adds sector fixed effects (ISIC Rev.4 section or own classifier, assigned to each occupation by its modal sector in the base year) interacted with the quarter, so that $\hat{\tau}^c$ is identified by comparing occupations of different exposure within the same sector and quarter, absorbing any sectoral recovery trajectory. Employment-weighted TWFE, SEs clustered by occupation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

and tertiary education (T_o), both predetermined. Three readings. First, the employment formality gradient survives the skill control where it was significant: Costa Rica (+0.476, $p < 0.01$), Ecuador (+0.314, $p < 0.10$), and Uruguay (−0.467, $p < 0.10$), with T_o close to zero in employment for all three; Brazilian and Bolivian formal employment, insignificant in the baseline, remain so here. Second, in wages both margins operate: the formality gradient persists in Ecuador (−0.313, $p < 0.01$), Costa Rica (−0.327, $p < 0.05$) and Brazil (−0.242, $p < 0.01$), but skill plays its own role with variable sign (negative in Mexico, −0.136, $p < 0.01$, and Costa Rica; positive in Uruguay, +0.153, $p < 0.01$; in Brazil T_o is precise but works only on hours, +1.312, $p < 0.01$, never the wage). Third, the exercise has clear limits: in Peru the occupational correlation between F_o and T_o (0.73) prevents separating the margins with precision, and Bolivia is the extreme case: with collinearity nearly as high as Peru’s

Table A5: Formal and informal channels under sector×quarter fixed effects

	Channel	log employment		log wage		hours	
		Baseline	Sec×t	Baseline	Sec×t	Baseline	Sec×t
Bolivia	$\hat{\tau}_F$	0.188	0.328**	-0.092**	-0.092*	0.180	1.463
	$\hat{\tau}_I$	-0.152	0.043	0.003	0.084	-3.393*	-1.462
Brazil	$\hat{\tau}_F$	0.167***	0.079	-0.069***	-0.042	0.080	-0.889**
	$\hat{\tau}_I$	-0.100	-0.181	0.202***	0.199**	0.421	1.113
Colombia	$\hat{\tau}_F$	0.111	0.191*	0.125***	0.090*	-2.714***	-2.345***
	$\hat{\tau}_I$	-0.313	-0.412	0.101**	0.087	1.401**	1.079
Costa Rica	$\hat{\tau}_F$	0.399***	0.204	-0.267***	-0.166**	-4.089***	-1.715
	$\hat{\tau}_I$	-0.082	-0.184	0.205	0.119	-4.185	-3.866
Ecuador	$\hat{\tau}_F$	-0.060	0.005	-0.208***	-0.205***	-0.628	-2.032**
	$\hat{\tau}_I$	-0.341***	-0.216*	0.193***	0.186**	-0.557	-1.155
Mexico	$\hat{\tau}_F$	0.106*	0.018	-0.007	-0.015	0.880***	0.843**
	$\hat{\tau}_I$	-0.045	-0.077	-0.044	-0.018	-1.136	-1.076
Peru	$\hat{\tau}_F$	0.406***	0.269**	-0.144***	-0.123**	-0.619	-3.101***
	$\hat{\tau}_I$	0.003	-0.188	0.009	0.063	1.636	0.969
Uruguay	$\hat{\tau}_F$	-0.179***	-0.207***	0.021	-0.013	0.139	-0.336
	$\hat{\tau}_I$	0.216	0.178	-0.130	-0.175	4.393*	4.507

Notes: “Baseline” reproduces Table 6; “Sec×t” adds modal-sector-by-quarter fixed effects as in Table A4, so that the channels are identified from within-sector variation. SEs clustered by occupation (omitted for space; significance shown by asterisks). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

(0.71), the formality wage gradient, significant in the baseline (-0.092 , $p < 0.05$), loses all significance against T_o (-0.123 , n.s.), so the horse race cannot pronounce on the Bolivian mechanism. Colombia is left out because its harmonized base drops the education variable; firm size is not harmonized across the eight surveys. The prudent conclusion is that formality is not a mere reflection of skill, though a skill margin does exist for wages; the individual-level decomposition, with each worker’s education and firm size, belongs to the project’s second paper, where Chatterji et al. (2025), with workplace chatbot use rising in education, makes individual education the first dimension.

Teleworkability is the other natural confounder: formal occupations are more teleworkable, and the post-COVID recovery differentially favored remote employment. Table A9 repeats the horse race with the teleworkability W_o of Dingel and Neiman (2020) in place of T_o , with two advantages: lower collinearity ($\text{corr}(F_o, W_o)$ between 0.20 and 0.41, against 0.44–0.73 with T_o) and all eight countries, Colombia included. The result is the battery’s most favorable for the formality margin: the formal employment gradient is significant at least at the 10 percent level in all eight countries, positive in seven (Costa Rica +0.490, $p < 0.01$; Peru +0.402, $p < 0.01$; Bolivia +0.402, $p < 0.05$; Ecuador +0.287, $p < 0.05$; Brazil +0.268, $p < 0.10$; Mexico +0.173, $p < 0.05$; Colombia +0.490, $p < 0.10$) and negative in Uruguay (-0.386 , $p < 0.10$), reproducing its reversal. The formal wage gradient retains

Table A6: Robustness of the formality definition: baseline (pensions, all employed) versus wage earners only

	Channel	log employment		log wage		hours	
		Baseline	Wage e.	Baseline	Wage e.	Baseline	Wage e.
Bolivia	$\hat{\tau}_F$	0.188	0.141	−0.092**	−0.077**	0.180	0.021
	$\hat{\tau}_I$	−0.152	−0.130	0.003	−0.006	−3.393*	−4.109**
Brazil	$\hat{\tau}_F$	0.167***	0.170***	−0.069***	−0.061***	0.080	0.240
	$\hat{\tau}_I$	−0.100	−0.182	0.202***	0.222***	0.421	0.290
Colombia	$\hat{\tau}_F$	0.111	0.008	0.125***	0.125***	−2.714***	−1.896***
	$\hat{\tau}_I$	−0.313	−0.314	0.101**	0.040	1.401**	3.406***
Costa Rica	$\hat{\tau}_F$	0.399***	0.357***	−0.267***	−0.223***	−4.089***	−4.139***
	$\hat{\tau}_I$	−0.082	−0.026	0.205	0.128	−4.185	−3.870
Ecuador	$\hat{\tau}_F$	−0.060	−0.136	−0.208***	−0.130***	−0.628	−1.009
	$\hat{\tau}_I$	−0.341***	−0.365*	0.193***	0.364***	−0.557	1.693
Mexico	$\hat{\tau}_F$	0.106*	0.104*	−0.007	−0.003	0.880***	0.996***
	$\hat{\tau}_I$	−0.045	−0.084	−0.044	−0.031	−1.136	−0.526
Peru	$\hat{\tau}_F$	0.406***	0.442***	−0.144***	−0.136***	−0.619	−1.166
	$\hat{\tau}_I$	0.003	−0.131	0.009	0.032	1.636	2.340*
Uruguay	$\hat{\tau}_F$	−0.179***	−0.165**	0.021	0.016	0.139	0.242
	$\hat{\tau}_I$	0.216	0.172	−0.130	−0.128	4.393*	9.045***

Notes: “Baseline” reproduces Table 6 (harmonized formality over all employed workers). “Wage e.” redefines the baseline formality share F_o using only the wage earners of each occupation (the wage earner’s pension contribution; the self-employed are left out of the computation of F_o), keeping the sample and the specification unchanged. SEs clustered by occupation (omitted for space; significance shown by asterisks). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

sign and significance in Costa Rica (−0.446, $p < 0.01$), Ecuador (−0.379, $p < 0.01$), Brazil (−0.274, $p < 0.01$), Bolivia (−0.177, $p < 0.01$), and Peru (−0.149, $p < 0.10$). Teleworkability plays its own role in isolated cells (the Mexican wage, −0.070, $p < 0.01$) but nowhere absorbs the formality margin; Bolivian formal employment, imprecise in the baseline and collinear in the education horse race, emerges significant here, as under the sector-by-quarter control. The scope of this reading deserves stating: the exercise lives inside the interaction specification, so the strength of the employment gradient is not evidence of a formal employment expansion, a margin that direct sector-specific estimation and the lower-order exercise closing this appendix confine to Peru and, marginally, Costa Rica. What the control sustains is that formality, not teleworkability, is the dimension organizing the heterogeneity.

A fifth exercise descends to the individual worker. The fall in the exposed formal wage admits a *price* story (incumbents lose income) and a *composition* story (the expansion absorbs entrants at lower entry wages and the average falls without anyone’s wage falling); Table A10 separates them by tenure, entrants (under twelve months) against incumbents. The contrast is conditional heterogeneity rather than a comparison of two fixed populations: tenure under

Table A7: Influence of major occupational groups: leave-one-out of the aggregate $\hat{\tau}^c$

	Baseline $\hat{\tau}^c$	Minimum (omitted group)	Maximum (omitted group)	Groups
<i>log employment</i>				
Bolivia	0.069	−0.046 (excl. 7)	0.161 (excl. 9)	9
Brazil	0.123***	0.088 (excl. 6)	0.154 (excl. 4)	9
Colombia	−0.017	−0.059 (excl. 4)	0.042 (excl. 5)	9
Costa Rica	0.302***	0.212 (excl. 9)	0.352 (excl. 3)	9
Ecuador	−0.164**	−0.212 (excl. 3)	−0.089 (excl. 5)	9
Mexico	0.064	0.030 (excl. 9)	0.088 (excl. 4)	9
Peru	0.283**	0.210 (excl. 9)	0.408 (excl. 5)	9
Uruguay	−0.163**	−0.187 (excl. 5)	−0.015 (excl. 4)	9
<i>log monthly wage</i>				
Bolivia	−0.059*	−0.080 (excl. 9)	−0.043 (excl. 2)	9
Brazil	−0.025	−0.052 (excl. 9)	−0.002 (excl. 4)	9
Colombia	0.118***	0.098 (excl. 2)	0.162 (excl. 4)	9
Costa Rica	−0.178***	−0.284 (excl. 9)	−0.115 (excl. 2)	9
Ecuador	−0.058	−0.114 (excl. 5)	−0.007 (excl. 2)	9
Mexico	−0.017	−0.038 (excl. 6)	−0.005 (excl. 9)	9
Peru	−0.097**	−0.136 (excl. 5)	−0.062 (excl. 9)	9
Uruguay	0.015	−0.011 (excl. 2)	0.046 (excl. 5)	9
<i>weekly hours</i>				
Bolivia	−1.067	−1.713 (excl. 9)	0.024 (excl. 5)	9
Brazil	0.135	0.013 (excl. 2)	0.275 (excl. 5)	9
Colombia	−1.468***	−1.645 (excl. 5)	−1.040 (excl. 2)	9
Costa Rica	−4.027***	−5.202 (excl. 4)	−2.836 (excl. 9)	9
Ecuador	−0.611	−0.991 (excl. 4)	0.087 (excl. 9)	9
Mexico	0.325	−0.001 (excl. 3)	0.680 (excl. 4)	9
Peru	0.135	−0.098 (excl. 6)	0.715 (excl. 3)	9
Uruguay	0.319	−0.171 (excl. 9)	0.534 (excl. 4)	9

Notes: Each row re-estimates the baseline specification omitting, one at a time, each major occupational group (first digit of the country's native classifier) and reports the range of the coefficient. A narrow range around the baseline $\hat{\tau}^c$ indicates that no single major group drives the result on its own. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ (from the baseline specification).

twelve months identifies workers recently arrived in their current job, who may be entrants to the labor market but also occupational switchers, and both the switching and the sector of arrival can themselves respond to the shock. What the exercise establishes is therefore that the wage decline is located among the recently hired and not among the established, which is what distinguishes the composition story from the price story; whether the recently hired are new to the market or new to the occupation is not identified with repeated cross-sections, and their observed wage may also reflect shifts in age or education within that group. The result is the sharpest regularity in the battery: the entrant-incumbent differential is negative and significant in the formal channel of the five countries with a tenure variable, and in the informal channel of all but Peru. In Peru the separation is complete: exposed formal incumbents' wages do not move (-0.009 , $p = 0.78$) and the entire effect falls on entry wages (-0.222 , $p < 0.01$); in Costa Rica the entrant effect (-0.204 , $p < 0.01$) is two and a half times the incumbent's (-0.082 , $p < 0.05$), and in Ecuador the incumbent effect (-0.210 ,

Table A8: Horse-race: baseline formality (F_o) versus tertiary-education share (T_o)

	Term	log employment	log monthly wage	weekly hours
Bolivia (corr $F_o, T_o = 0.71$)	$\beta \times \text{post}$	-0.136 (0.185)	-0.008 (0.071)	- 4.006* (2.072)
	$\times F_o$	0.378 (0.283)	-0.123 (0.119)	2.095 (3.187)
	$\times T_o$	-0.050 (0.318)	0.037 (0.134)	1.914 (3.324)
Brazil (corr $F_o, T_o = 0.44$)	$\beta \times \text{post}$	-0.123 (0.118)	0.212*** (0.069)	0.160 (0.845)
	$\times F_o$	0.202 (0.159)	- 0.242*** (0.085)	-1.078 (0.945)
	$\times T_o$	0.116 (0.071)	-0.052 (0.035)	1.312*** (0.427)
Costa Rica (corr $F_o, T_o = 0.49$)	$\beta \times \text{post}$	-0.084 (0.162)	0.265** (0.129)	-3.004 (2.759)
	$\times F_o$	0.476*** (0.176)	- 0.327** (0.163)	2.935 (3.303)
	$\times T_o$	0.009 (0.183)	- 0.282** (0.123)	-5.533 (3.616)
Ecuador (corr $F_o, T_o = 0.55$)	$\beta \times \text{post}$	- 0.325** (0.130)	0.235*** (0.068)	-2.420 (1.654)
	$\times F_o$	0.314* (0.175)	- 0.313*** (0.075)	- 3.939*** (1.429)
	$\times T_o$	-0.052 (0.178)	- 0.137** (0.068)	6.039*** (1.143)
Mexico (corr $F_o, T_o = 0.55$)	$\beta \times \text{post}$	-0.040 (0.071)	-0.034 (0.038)	-1.147 (1.204)
	$\times F_o$	0.208* (0.119)	0.142*** (0.046)	1.899 (1.171)
	$\times T_o$	-0.074 (0.092)	- 0.136*** (0.039)	0.153 (0.634)
Peru (corr $F_o, T_o = 0.73$)	$\beta \times \text{post}$	-0.042 (0.180)	-0.007 (0.109)	0.649 (1.124)
	$\times F_o$	0.284 (0.195)	-0.196 (0.186)	- 4.814** (1.946)
	$\times T_o$	0.167 (0.208)	0.060 (0.201)	3.604 (2.251)
Uruguay (corr $F_o, T_o = 0.47$)	$\beta \times \text{post}$	0.224 (0.185)	-0.117 (0.151)	4.440* (2.444)
	$\times F_o$	- 0.467* (0.242)	0.037 (0.158)	- 4.664* (2.560)
	$\times T_o$	0.095 (0.123)	0.153*** (0.055)	0.547 (1.087)

Notes: Specification $Y = \tau_0 \beta \text{post} + \tau_F \beta \text{post} F_o + \tau_T \beta \text{post} T_o$ with occupation and quarter fixed effects, where T_o is the occupation's share of workers with tertiary education in the base year. A τ_F that survives the control for T_o indicates that the formality channel is not a mere reflection of skill. Colombia is left out: its harmonized database does not retain the education variable. The country-specific education mappings are documented in the replication package and are subject to verification. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

$p < 0.01$) is half of what entrants bear (-0.417 , $p < 0.01$), so in those two countries the adjustment is only partly compositional. Brazil replicates the separation with the sample's largest magnitude: formal incumbents' wages do not move ($+0.026$, n.s.) while entrants' fall by -0.356 ($p < 0.01$), an entry compression larger than Peru's; its informal channel adds the country's nuance, incumbents gaining ($+0.119$, $p < 0.01$) and entrants losing without significance (-0.117 , n.s.), so the aggregate informal wage spillover of Table 6 is, individually, a recomposition toward established informal workers and not a generalized improvement. In Colombia, symmetrically, the wage gains concentrate among incumbents ($+0.198$, $p < 0.01$) and never reach entrants. The mechanism behind the central finding is thus identified, and it is composition rather than price, though in pure form only in two countries: the fall in the exposed formal wage always concentrates in entry wages, and the price of incumbents' labor holds in Peru and Brazil while it partly gives way in Costa Rica and Ecuador: complementarity and supply pressure at entry, hard to reconcile with generalized erosion of bargaining power among incumbents. What the exercise identifies is the wage channel, not pattern P's quantity half, which does not rest on it. The concentration among entrants coincides

Table A9: Horse-race: baseline formality (F_o) versus teleworkability (W_o)

	Term	log employment	log monthly wage	weekly hours
Bolivia (corr $F_o, W_o = 0.41$)	$\beta \times \text{post}$	-0.101 (0.164)	-0.063 (0.056)	- 4.344 ** (1.885)
	$\times F_o$	0.401 ** (0.164)	- 0.177 *** (0.061)	2.382 (1.925)
	$\times W_o$	-0.119 (0.160)	0.155 *** (0.046)	2.257 * (1.171)
Brazil (corr $F_o, W_o = 0.20$)	$\beta \times \text{post}$	-0.098 (0.126)	0.195 ** (0.079)	0.758 (0.872)
	$\times F_o$	0.268 * (0.152)	- 0.274 *** (0.079)	-0.219 (0.966)
	$\times W_o$	-0.002 (0.053)	0.010 (0.030)	-0.503 (0.327)
Colombia (corr $F_o, W_o = 0.28$)	$\beta \times \text{post}$	-0.105 (0.183)	0.049 (0.059)	1.106 * (0.635)
	$\times F_o$	0.490 * (0.289)	0.008 (0.063)	- 4.211 *** (0.775)
	$\times W_o$	- 0.279 * (0.148)	0.071 * (0.042)	0.402 (0.475)
Costa Rica (corr $F_o, W_o = 0.34$)	$\beta \times \text{post}$	-0.087 (0.182)	0.299 ** (0.128)	-3.382 (2.889)
	$\times F_o$	0.490 *** (0.144)	- 0.446 *** (0.135)	0.246 (2.897)
	$\times W_o$	-0.015 (0.133)	-0.138 (0.091)	-1.035 (2.121)
Ecuador (corr $F_o, W_o = 0.27$)	$\beta \times \text{post}$	- 0.316 ** (0.148)	0.259 *** (0.076)	-0.360 (1.410)
	$\times F_o$	0.287 ** (0.119)	- 0.379 *** (0.051)	0.009 (1.819)
	$\times W_o$	-0.032 (0.102)	- 0.095 * (0.056)	-0.314 (1.366)
Mexico (corr $F_o, W_o = 0.29$)	$\beta \times \text{post}$	-0.007 (0.069)	-0.000 (0.032)	-0.841 (1.125)
	$\times F_o$	0.173 ** (0.085)	0.061 * (0.032)	2.171 * (1.167)
	$\times W_o$	-0.064 (0.052)	- 0.070 *** (0.017)	-0.465 (0.680)
Peru (corr $F_o, W_o = 0.39$)	$\beta \times \text{post}$	0.081 (0.213)	- 0.178 * (0.096)	-0.929 (1.945)
	$\times F_o$	0.402 *** (0.145)	- 0.149 * (0.087)	-2.207 (1.475)
	$\times W_o$	-0.073 (0.124)	0.178 ** (0.069)	2.434 * (1.329)
Uruguay (corr $F_o, W_o = 0.23$)	$\beta \times \text{post}$	0.279 (0.207)	-0.120 (0.143)	4.426 * (2.455)
	$\times F_o$	- 0.386 * (0.202)	0.153 (0.141)	- 4.248 * (2.259)
	$\times W_o$	-0.077 (0.091)	-0.013 (0.044)	-0.033 (0.925)

Notes: Specification $Y = \tau_0 \beta \text{post} + \tau_F \beta \text{post} F_o + \tau_W \beta \text{post} W_o$ with occupation and quarter fixed effects, where W_o is the occupation's teleworkability according to Dingel and Neiman (2020) (O*NET-SOC classification, mapped to the native occupation through the same crosswalk chain as the treatment; coverage above 99 percent in the eight countries). A τ_F that survives controlling for W_o indicates that the formality channel does not reflect the differential recovery of teleworkable employment. Unlike the educational horse-race (Table A8), Colombia enters this exercise. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

with recent US payroll and UK vacancy evidence (Brynjolfsson, Chandar, & Chen, 2025; Klein Teeselink, 2025), with an instructive difference: there the adjustment is documented on the quantity margin (fewer entry hires), and here a price margin at entry is added to it. The quantity moves in the region too, and in the same direction: the formal channel's entrant share falls significantly in four of the five countries (Costa Rica -0.080 , Colombia -0.045 , Brazil -0.028 in the formal channel and -0.059 in the informal, both $p < 0.01$ and Ecuador -0.020), with Peru the only exception ($+0.019$, n.s.), so the adjustment comes from the hiring price rather than from higher turnover; and in Colombia that fall in formal entry has its counterpart in informal entry ($+0.062$, $p < 0.05$), an echo of the theoretical framework's reallocation channel. Mexico and Uruguay are left out because the ENOE does not collect tenure and the redesigned ECH stopped publishing it, and Bolivia because the ECE dropped the tenure question after 2020Q2, before the study window; the Ecuadorian

Table A10: Mechanism of the wage pattern: entrants versus incumbents (individual worker level)

		log monthly wage			Entrant share
Country	Employment	Incumbents	Entrants	Differential	(entry margin)
Bolivia	<i>N/A (see notes)</i>				
Brazil	Formal	0.026 (0.020)	− 0.356*** (0.051)	− 0.382***	− 0.028***
	Informal	0.119*** (0.037)	−0.117 (0.091)	− 0.236**	− 0.059***
Colombia	Formal	0.198*** (0.038)	−0.054 (0.041)	− 0.252***	− 0.045***
	Informal	0.226*** (0.051)	−0.089 (0.063)	− 0.315***	0.062**
Costa Rica	Formal	− 0.082** (0.039)	− 0.204*** (0.063)	− 0.122**	− 0.080***
	Informal	−0.061 (0.130)	− 0.297** (0.141)	− 0.236*	0.033
Ecuador	Formal	− 0.210*** (0.038)	− 0.417*** (0.063)	− 0.207***	− 0.020**
	Informal	0.109** (0.044)	−0.173 (0.130)	− 0.282**	−0.025
Mexico	<i>N/A (see notes)</i>				
Peru	Formal	−0.009 (0.031)	− 0.222*** (0.055)	− 0.213***	0.019
	Informal	−0.104 (0.076)	0.180 (0.156)	0.284	0.036
Uruguay	<i>N/A (see notes)</i>				

Notes: Estimates at the WORKER LEVEL on employed workers with wages from the same surveys as the main panel, 2021Q1–2025Q4. Entrant: less than twelve months in the current job (Colombia p6426; Costa Rica e3/d5; Ecuador p45; Peru p513a1; Brazil V4040); incumbent: twelve months or more. The wage columns report the coefficient on $\beta_o \times \text{post}_t$ for incumbents, the total effect for entrants, and their differential, from the individual-level regression with occupation and quarter fixed effects, entrant status and its interaction with post as controls, weighted by the expansion factor and with errors clustered by occupation, estimated separately for formal and informal employment (harmonized individual definition). The last column is the effect of $\beta_o \times \text{post}_t$ on the entrant share of the group’s occupation×quarter cell. Mexico and Uruguay are excluded because the ENOE does not collect tenure in the current job and the redesigned ECH stopped publishing it; Bolivia published tenure in the ECE but the question was discontinued after 2020Q2, with no coverage in the study window. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

pre-trends impose, as elsewhere, additional caution on its row.

The sixth exercise answers the pre-trends diagnostic with an estimator that incorporates them by construction: the synthetic DiD of Arkhangelsky et al. (2021) reweights control occupations and pre-treatment periods so the synthetic control replicates the treated group’s prior trajectory, attacking the design’s weak flank (the wage pre-trends). Admitting no continuous dose, it runs on the binary high/low β split of the Callaway–Sant’Anna specification, over each country’s balanced panel of occupations. Table A11 reports a remarkable result: the two original pattern P countries recover their full signature in the binary aggregate, employment rising and the wage falling in Costa Rica (+0.150, SE 0.049; −0.060, SE 0.034) and Peru (+0.108, SE 0.038; −0.074, SE 0.032), and Mexico echoes it more weakly (+0.039 and −0.022, both with t -ratios close to 2). The negative aggregates of Ecuador and Uruguay, in contrast, do not survive the reweighting (+0.022 and −0.012, not significant), consistent with the signal in those countries residing in the informal channel and the reversal of the formal channel, not in the average. Brazil partially confirms its signature here: employment

Table A11: Synthetic DiD (2021): binary high/low β split, balanced panel

	log employment	log monthly wage	weekly hours	T/C	Emp. cov.
Bolivia	0.011 (0.042)	−0.001 (0.023)	0.041 (0.581)	59/35	99%
Brazil	0.049** (0.020)	−0.013 (0.009)	−0.012 (0.108)	203/142	100%
Colombia	−0.035 (0.036)	−0.024 (0.017)	−0.270 (0.197)	173/115	99%
Costa Rica	0.150*** (0.049)	− 0.060* (0.034)	− 1.149* (0.588)	46/45	92%
Ecuador	0.022 (0.029)	−0.028 (0.017)	−0.086 (0.240)	159/63	98%
Mexico	0.039** (0.019)	− 0.022** (0.010)	−0.026 (0.182)	167/159	100%
Peru	0.108*** (0.038)	− 0.074** (0.032)	1.047* (0.561)	63/32	95%
Uruguay	−0.012 (0.075)	0.006 (0.038)	−0.091 (0.654)	39/46	89%

Notes: Estimator of Arkhangelsky et al. (2021) on the binary split at the base-year employment-weighted median of β (the same split as the Callaway–Sant’Anna specification), balanced panel of occupations without the 2022Q4 buffer and without employment weighting (the estimator weights units and periods endogenously to replicate the pre-treatment trajectory of the treated group). “T/C” reports treated and control occupations; “Emp. cov.” the share of total employment covered by the balanced subsample. Placebo standard errors from the estimator itself. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

remains positive and significant (+0.049, $p < 0.05$, SE 0.020), though smaller than in the aggregate TWFE, while the wage does not reach significance (−0.013, SE 0.009), unsurprising since it did not in the baseline specification either. Bolivia, in contrast, shows no signal under SDID in any outcome (+0.011 in employment, −0.001 in wages), and its own placebo (Section 5.7) suggests the estimator is imprecise in its panel, the smallest of the eight. Overall, the pre-trend-robust estimator sustains the binary signature where pattern P is the country’s reading, without raising it above its exploratory status, with Brazil contributing the sharpest employment signal among the two new countries; Honest-DiD remains the preregistered primary criterion, the SDID triangulation rather than replacement.

The seventh exercise attacks the relation between estimator and estimand. As Section 3.3 makes precise, the continuous-dose TWFE coefficient weights causal responses in ways that need not match the dose density and are sensitive to the mass of unexposed occupations (Callaway, Goodman-Bacon, & Sant’Anna, 2025). Table A12 estimates those authors’ recommended summary, $ACRT^{\text{glob}}$: the average derivative of the long difference’s dose-response curve, fitted by flexible regression among the treated occupations and aggregated with the dose distribution’s own weights. Two readings. First, and central: *all eleven significant aggregate estimates retain their sign under the two flexible specifications*, with comparable magnitudes where the pattern is sharpest (Peruvian employment +0.284 and +0.293 against +0.283 from the TWFE; Costa Rican +0.197 and +0.247 against +0.302), so the implicit TWFE weighting fabricates no sign. Second, and in honesty: the flexible estimation on the collapsed panel loses precision (bootstrap standard errors 1.5 to 3 times the TWFE’s) and most cells lose individual significance; those that retain it are Costa Rican and Colombian hours, the Colombian wage, and Brazilian employment under the cubic polynomial, and the only cell gaining marginal significance where the TWFE had none is the Mexican wage (−0.076 and −0.047, $p < 0.10$), compatible with noise across 24 contrasts. The TWFE remains primary for precision and comparability with the literature; the $ACRT^{\text{glob}}$ bounds

Table A12: From TWFE to $ACRT^{\text{glob}}$: causal summary with dose-density weights (2025)

	TWFE (baseline)	$ACRT^{\text{glob}}$ B-splines	$ACRT^{\text{glob}}$ cubic
<i>log employment</i>			
Bolivia	0.069 (0.121)	−0.131 (0.351)	0.022 (0.160)
Brazil	0.123*** (0.042)	0.016 (0.143)	0.132** (0.060)
Colombia	−0.017 (0.089)	−0.140 (0.220)	0.014 (0.130)
Costa Rica	0.302*** (0.098)	0.197 (0.249)	0.247 (0.164)
Ecuador	− 0.164** (0.082)	−0.208 (0.303)	−0.115 (0.152)
Mexico	0.064 (0.049)	0.003 (0.099)	0.057 (0.058)
Peru	0.283** (0.112)	0.284 (0.314)	0.293 (0.188)
Uruguay	− 0.163** (0.073)	−0.046 (0.140)	−0.104 (0.070)
<i>log monthly wage</i>			
Bolivia	− 0.059* (0.030)	−0.042 (0.071)	−0.049 (0.042)
Brazil	−0.025 (0.025)	0.007 (0.042)	0.005 (0.027)
Colombia	0.118*** (0.033)	0.104 (0.094)	0.094** (0.043)
Costa Rica	− 0.178*** (0.050)	−0.081 (0.182)	−0.089 (0.101)
Ecuador	−0.058 (0.041)	−0.046 (0.098)	0.018 (0.068)
Mexico	−0.017 (0.024)	− 0.076* (0.041)	− 0.047* (0.026)
Peru	− 0.097** (0.049)	−0.079 (0.149)	− 0.139* (0.078)
Uruguay	0.015 (0.027)	−0.033 (0.094)	0.010 (0.045)
<i>weekly hours</i>			
Bolivia	−1.067 (1.153)	−0.743 (1.832)	−1.439 (1.102)
Brazil	0.135 (0.173)	−0.243 (0.522)	0.261 (0.263)
Colombia	− 1.468*** (0.356)	−0.958 (0.642)	− 1.168*** (0.384)
Costa Rica	− 4.027*** (1.130)	− 5.214** (2.347)	− 4.449*** (1.487)
Ecuador	−0.611 (0.741)	−0.894 (2.272)	−1.365 (1.271)
Mexico	0.325 (0.478)	− 1.078* (0.591)	−0.448 (0.476)
Peru	0.135 (0.675)	−0.373 (3.915)	0.120 (1.638)
Uruguay	0.319 (0.451)	−1.338 (2.102)	0.573 (0.626)

Notes: The TWFE column reproduces Table 5. $ACRT^{\text{glob}}$: estimator of Callaway, Goodman-Bacon, and Sant’Anna (2025) described in the text (long difference by occupation, flexible adjustment among treated units with cubic B-splines or a global cubic polynomial, and derivative averaged with dose-density and baseline-employment weights). SEs from an occupation bootstrap (999 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

what the choice of weights can and cannot generate.

The eighth exercise examines differential seasonality by exposure group. Formal payment calendars (legislated mid-year and end-of-year bonuses) and informal demand peaks (Christmas, commercial dates) generate seasonal income patterns that differ between exposed and unexposed occupations and that quarter fixed effects do not absorb; the seasonal component of the between-group wage gap reaches 2 to 5 percentage points of log wage in Colombia, Ecuador, Uruguay, and Peru. Re-estimating with occupation-by-calendar-quarter fixed effects, which absorb each occupation’s own seasonal pattern, leaves two readings (the full table, entirely null, is in the replication package). First, the estimates are insensitive to the control: the employment and wage $\hat{\tau}^c$ barely move in the third decimal (Costa Rica +0.302

to +0.306; Peru +0.283 to +0.281; Brazil +0.123 to +0.125), and those for hours, noisier, move more without changing sign or significance, so differential seasonality does not contaminate the results. Second, and in honesty, the seasonal control does not rehabilitate the wage pre-trends: absorbing the seasonal noise, the Wald test gains precision and rejects wage parallelism with equal or greater force, so those violations reflect low-frequency drift, not calendar artifacts, which reinforces treating the Honest-DiD bounds as the primary criterion.

The ninth exercise examines the weighting. The baseline weights each cell by its contemporaneous expanded employment, the very object the treatment acts upon: if exposed employment expands, exposed cells weigh more after the shock by construction. Re-estimating the two channels with each cell weighted by its occupation’s average pre-period employment, fixed over time and predetermined with respect to the launch, leaves the picture intact: all the significant coefficients of [Tables 5 and 6](#) retain sign and significance with very close magnitudes (Costa Rican formal employment +0.399 to +0.380, the Peruvian +0.406 to +0.413, the Brazilian +0.167 to +0.164), and the only estimates changing category gain precision (the Bolivian wage from $p < 0.10$ to $p < 0.05$). The full table, with aggregates and channels, is in the replication package. Contemporaneous weighting therefore fabricates none of the paper’s patterns.

A final exercise examines an omission in the heterogeneity specification itself. [equation \(9\)](#) includes the dose interacted with baseline formality but not the lower-order term $F_o \times \text{post}$. Without it, any post-2022 shift associated with the occupation’s formality rather than with exposure can load onto the two interactions of interest, all the more so because β_o and F_o are correlated across occupations. [Table A13](#) re-estimates the formal channel including that term. The verdict is uncomfortable and is best stated plainly: the omitted interaction is significant in Costa Rica, Brazil, Peru and Uruguay, and once it is included the formal employment coefficient collapses in the three pattern P countries, from +0.399 to +0.009 in Costa Rica, from +0.406 to +0.068 in Peru and from +0.167 to -0.040 in Brazil. The formal wage is not spared, and it is worth saying so just as plainly: it too is absorbed in Costa Rica (from -0.267 to $+0.030$), Peru (from -0.144 to $+0.012$) and Bolivia (from -0.092 to -0.045), so within the interaction specification it survives only in Brazil (-0.092 , $p < 0.01$) and Ecuador (-0.229 , $p < 0.01$). This is not a precision problem: standard errors inflate by factors of 1.0 to 2.4 while the point estimates move to zero, and in Brazil the standard error does not move at all. The honest reading is that, under this specification, an exposure effect concentrated in formal occupations cannot be separated from a formality-specific post-2022 shift that does not vary with the dose. The sector-specific estimates of [Section 5.1](#) are the answer to this limitation, since they measure each sector’s own outcome without relying on the interaction, and there the formal wage channel survives in six of eight countries, five of them with a clean sector-specific placebo, while the formal employment expansion is reduced to Peru and, marginally, Costa Rica.

Table A13: Formal channel with and without the lower-order interaction $F_o \times \text{post}$

Country	Employment (log)			Monthly wage (log)		
	Baseline	With $F_o \times \text{post}$	$F_o \times \text{post}$	Baseline	With $F_o \times \text{post}$	$F_o \times \text{post}$
Bolivia	+0.188	+0.224	−0.023	− 0.092 **	−0.045	−0.029
Brazil	+ 0.167 ***	−0.040	+ 0.186 ***	− 0.069 ***	− 0.092 ***	+0.021
Colombia	+0.111	−0.039	+0.102	+ 0.125 ***	+0.055	+0.048
Costa Rica	+ 0.399 ***	+0.009	+ 0.387 ***	− 0.267 ***	+0.030	− 0.294 ***
Ecuador	−0.060	−0.161	+0.058	− 0.208 ***	− 0.229 ***	+0.012
Mexico	+0.106*	+0.007	+0.068	−0.007	− 0.066 **	+0.040
Peru	+ 0.406 ***	+0.068	+ 0.226 **	− 0.144 ***	+0.012	− 0.104 **
Uruguay	− 0.179 ***	− 0.283 ***	+ 0.166 **	+0.021	+0.050	−0.046

Notes: Formal-channel coefficient $\hat{\tau}_F^c$ under the specification of Table 6 and under the same specification augmented with $F_o \times \text{post}$, the lower-order interaction the baseline omits. The third column of each block reports that interaction’s own coefficient. Occupation and quarter fixed effects, employment weighting and occupation-clustered standard errors in both specifications. Coefficients significant at 5 percent in bold. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A14: Formal test of cross-country heterogeneity

Block	Outcome	Range of $\hat{\tau}^c$	Common effect	Q	p	I^2 (%)
Aggregate	Employment (log)	[−0.164, +0.302]	+0.058	30.0***	0.000	77
Aggregate	Monthly wage (log)	[−0.178, +0.118]	−0.018	34.6***	0.000	80
Aggregate	Weekly hours	[−4.027, +0.325]	−0.158	31.5***	0.000	78
Formal channel	Employment (log)	[−0.179, +0.406]	+0.106	44.4***	0.000	84
Formal channel	Monthly wage (log)	[−0.267, +0.125]	−0.052	63.1***	0.000	89
Formal channel	Weekly hours	[−4.089, +0.880]	−0.422	81.9***	0.000	91
Informal channel	Employment (log)	[−0.341, +0.216]	−0.101	9.1	0.247	23
Informal channel	Monthly wage (log)	[−0.130, +0.205]	+0.057	21.8***	0.003	68
Informal channel	Weekly hours	[−4.185, +4.393]	+0.506	15.4**	0.032	54

Notes: Cochran’s Q test on the eight country estimates, which are independent because they come from separate surveys. Under the null of a common effect, $Q = \sum_c w_c (\hat{\tau}^c - \bar{\tau})^2$ with $w_c = 1/\hat{\sigma}_c^2$ is χ^2 with seven degrees of freedom. “Common effect” is the precision-weighted mean $\bar{\tau}$. I^2 is the share of observed dispersion exceeding sampling noise. Rejecting the null indicates that the cross-country differences exceed estimation uncertainty. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.